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Classification of all solutions to a third order ODE arising from the 2D laminar flow

Mingfeng Zhao [a], Changfeng Gui [b,*], Hongxia Guo [c], Ping Lin [d]

[a] *School of Mathematics, Southwestern University of Finance and Economics, Chengdu 611130, China*
[b] *Department of Mathematics, University of Macau, Taipa, Macau SAR 519000, China*
[c] *School of Mathematical Sciences, and Institute of Mathematics and Interdisciplinary Sciences, Tianjin Normal University, Tianjin 300387, China*
[d] *Division of Mathematics, University of Dundee, Dundee, DD1 4HN, UK*

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Abstract

This paper presents a complete classification of the bounded and unbounded solutions of a third-order nonlinear ordinary differential equation derived from two-dimensional laminar flow in a channel. Through careful analysis, we characterize all possible solution types under various conditions on initial data, providing an explicit description of their graphical behaviors. In particular, we prove that the only bounded solutions on the whole line are constant functions.
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* Corresponding author.
E-mail addresses: mingfengzhao@swufe.edu.cn (M. Zhao), changfenggui@um.edu.mo (C. Gui), hxguo@tjnu.edu.cn (H. Guo), p.lin@dundee.ac.uk (P. Lin).

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1. Introduction

Laminar flow through porous channels is a central topic in fluid mechanics. Its study is driven by its relevance to numerous applications, including filtration, membrane transport, transpiration cooling, and micro-circulation, where internal flows involve mass injection or suction. A classical framework was established by Berman [2], who demonstrated that the two-dimensional incompressible Navier–Stokes equations governing such flows can be reduced in many cases, via appropriate similarity transformations, to higher-order nonlinear ordinary differential equations (ODEs). These ODEs capture essential viscous phenomena—such as boundary layer formation, flow reversal, internal turning-point structures, and multiplicity of steady states—thereby providing a tractable framework for rigorous mathematical analysis.

Recent studies continue to employ Berman-type similarity reductions as a foundation for investigating more complex flow phenomena in porous configurations, including unsteady and wave–mean flow interactions, highlighting the enduring relevance of such reduced models [11].

For decades, this family of similarity solutions has served as a fundamental test case, driving continued analytical inquiry. The development of early asymptotic approaches was pioneered by Terrill and Shrestha [14,15], while subsequent contributions from Raithby [12], Robinson [13], later Zaturska et al. [16] and Lu [7–9], together with MacGillivray and Lu [10] and Cox and King [3,4], provided more refined analyses. These works uncovered multiple solution branches and increasingly intricate internal flow structures, thereby establishing the nonlinear ODEs derived from the similarity-reduced Navier–Stokes equations as a topic of lasting importance in applied analysis and asymptotic theory.

In our previous work [5], we investigated steady asymmetric laminar flow in a semi-infinite porous channel, a configuration that allows us to neglect boundary effects from the right-end opening. The upper and lower channel walls, positioned at $\tilde{x} = h_w$ and $\tilde{x} = -h_w$ respectively, are subject to unequal suction and injection velocities. The flow is governed by the following steady incompressible Navier–Stokes equations

$$\nabla \cdot \mathbf{V} = 0, \tag{1.1}$$

$$(\mathbf{V} \cdot \nabla)\mathbf{V} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{V}, \tag{1.2}$$

subject to the boundary conditions

$$u(0, y) = 0 \tag{1.3}$$

$$u(\tilde{x}, -h_w) = 0, \qquad v(\tilde{x}, -h_w) = -v_1, \tag{1.4}$$

$$u(\tilde{x}, h_w) = 0, \qquad v(\tilde{x}, h_w) = -v_2, \tag{1.5}$$

where $\mathbf{V} = (u(\tilde{x}, \tilde{y}), v(\tilde{x}, \tilde{y}))$ represents the velocity, p the pressure, ρ the density and ν the kinematic viscosity of the fluid, respectively. The flow is driven by a uniform injection through the upper wall of the channel with velocity $(0, -v_2)$ and a uniform suction through the lower wall with velocity $(0, -v_1)$.

A similarity solution for the channel flow is derived by introducing a stream function of the form

$$\psi = \nu x F(y), \tag{1.6}$$

where $x = \tilde{x}/h_w$ and $y = \tilde{y}/h_w$, and the corresponding velocity is

$$u = \frac{\nu x}{h_w} F'(y), \qquad v = -\frac{\nu}{h_w} F(y), \tag{1.7}$$

which leads to a third-order nonlinear ODE featuring a single undetermined integral constant K:

$$f''' + Re\left(ff'' - f'^2\right) = K, \tag{1.8}$$

with asymmetric boundary conditions

$$f(-1) = a, \quad f'(-1) = 0, \qquad f(1) = 1, \quad f'(1) = 0, \tag{1.9}$$

where $Re = h_w v_2/\nu$ denotes the cross-flow Reynolds number, $a = v_1/v_2 > 0$ and $f(y) = F(y)/Re$.

In [5] we have established both numerically and analytically that the boundary value problem described by (1.8) and (1.9) admits three distinct classes of solutions, designated as type I, type II and type III, across a broad range of Reynolds numbers. While this classification clarifies the global solution structure, a rigorous mathematical understanding of the underlying reduced equations, especially in singular perturbation regimes, remains largely incomplete. From a mathematical perspective, recent progress has been made toward establishing existence, uniqueness and approximation properties for similarity solutions of laminar flows in porous channels. In particular, Almuthaybiri and Tisdell [1] developed a fixed-point framework for a class of higher-order boundary value problems arising from porous channel flow models, yielding rigorous results on solvability and iterative approximation. Related analyses have also been carried out for modified boundary conditions, such as porous channel flows with wall slip [6]. However, these studies address different governing equations and do not consider the singular perturbation structure and nonlinear boundary-layer equations that emerge in the high Reynolds number regime studied here.

At large Reynolds numbers (with $\varepsilon = 1/Re$), each branch contains thin boundary layers characterized by different thickness scalings:

- **Type I**: A classical viscous boundary layer of thickness $O(\varepsilon)$ develops near the suction wall. The outer solution matches with the boundary layer in a standard fashion.
- **Type II**: Two interior turning points arise, forming an internal layer of width $O(\varepsilon|\log\varepsilon|)$. Matching the outer solution with both the interior layer and the $O(\varepsilon)$ wall layer requires controlling exponentially small terms and accurately locating the turning-points — a process notably more intricate than the type I construction.
- **Type III**: This branch displays the most complex multiscale structure, comprising an inviscid core, a transition region, and a thin but essential viscous layer adjacent to the suction wall. The outer solution satisfies only two of the four boundary conditions; the remaining two conditions must be recovered through the boundary-layer analysis. Consequently, the asymptotic solution construction reduces to solving a nonlinear boundary-layer ODE (to be explained below), the detailed properties of which were not analyzed in [5].

For the type III branch, introducing a stretched variable near the boundary $y = -1$

$$\eta = \frac{1+y}{\varepsilon},$$

yields the following asymptotic expansion with boundary layer correction terms (functions of η)

$$f(y) = f_0(y) + h_0(\eta) + \varepsilon\big(f_1(y) + h_1(\eta)\big) + \cdots. \tag{1.10}$$

Substituting it into (1.8) gives the reduced outer equation

$$f_0 f_0'' - (f_0')^2 = k_0. \tag{1.11}$$

At the boundary-layer scale, the leading-order profile $h_0(\eta)$ satisfies a third-order nonlinear equation

$$h_0''' + (h_0 + \beta)h_0'' - (h_0')^2 = 0, \qquad \beta = f_0(-1), \tag{1.12}$$

with

$$h_0(0) = a - \beta, \qquad h_0'(0) = 0,$$

and a decay condition as $\eta \to +\infty$ ensuring matching it with the outer solution.

Equation (1.12) captures the leading-order viscous behavior in the type III boundary layer. In our earlier work [5], the existence of a boundary layer solution to the third-order nonlinear boundary value problem was assumed without rigorous justification. Key aspects such as solvability, qualitative behavior, and admissible boundary-layer structures remained unexamined.

This paper is devoted to a mathematical analysis of the nonlinear boundary value problem of the ODE (1.12). Our analysis not only provides the missing theoretical justification for the type III asymptotic construction in porous channel flows, but also establishes a general analytical framework for a class of third-order nonlinear ODEs arising from similarity reductions of Navier–Stokes equations. The problem presents several distinctive and challenging features:

(1) It exhibits a strongly nonlinear, non-variational structure involving terms of form $hh'' - (h')^2$ — a form frequently arising from Navier-Stokes equations whose solution has a similarity feature, yet remains poorly understood from an analytical standpoint.
(2) Its solutions govern the viscous behavior required for the existence of the type III solution branch and the construction of its asymptotic expansion as outlined earlier.
(3) Unlike type I and type II boundary layers, the parameter $\beta = f_0(-1)$ must be determined concurrently with the solution of (1.12), resulting in a strongly coupled matching problem.
(4) To the best of our knowledge, no rigorous analytical results are available for this class of nonlinear ODEs of the form (1.12), and existing theories for third-order nonlinear ODEs do not directly apply.

Given the lack of rigorous analytical results, this work aims to develop a systematic framework for the equation (1.12), more precisely, our main result, **Theorem 1.1**, provides a complete

classification of the bounded and unbounded solutions to the equation (1.12). This effort fills a significant gap in the earlier asymptotic description and advances the mathematical understanding of flows with similarity behaviors under asymmetric permeability conditions.

Next we focus on the analysis of the nonlinear ODE (1.12). For the simplicity of notation we will write h_0 as h. For a given constant $\beta \in \mathbb{R}$, let's consider the following nonlinear ordinary differential equation:

$$h''' + (h+\beta)h'' = (h')^2. \tag{1}$$

Given any initial data $h(0)$, $h'(0)$ and $h''(0)$, we are going to classify all bounded and unbounded solutions of the equation (1) with the maximal solution intervals. It is evident from the equation (1) that

(A) any constant function is a bounded solution of the equation (1) on $\mathbb{R}$.
(B) if $\beta \neq 0$, for any constant $a \in \mathbb{R}\backslash\{0\}$, the function $y_1(x) := ae^{-\beta x}$ is a non-constant solution of the equation (1) on $\mathbb{R}$
(C) if $\beta = 0$, for any constant $a \in \mathbb{R}\backslash\{0\}$, the function $y_2(x) := \frac{6}{x+a}$ is a solution of the equation (1) on the interval $(-\infty, -a)$ or $(-a, \infty)$.
(D) if $y(x)$ is a solution of the equation (1), then for any $\alpha \in \mathbb{R}$, the vertical translating function $y(x) + \alpha$ is a solution of $h''' + (h + \beta - \alpha)h'' = (h')^2$.
(E) if $y(x)$ is a solution of the equation (1), then for any $a \in \mathbb{R}$, the horizontal translating function $y(x + a)$ is also a solution of the equation (1).
(F) if $\beta = 0$ and $y(x)$ is a solution of the equation (1), then for any $\lambda > 0$, the scaling function $\lambda y(\lambda x)$ is also a solution of the equation (1).
(G) if $\beta = 0$ and $y(x)$ is a solution of the equation (1), then the reflecting function $-y(-x)$ is also a solution of the equation (1).

In order to classify all bounded and unbounded solutions of equation (1), by considering the function $h(x) + \beta$, it suffices to classify all solutions of the following nonlinear ordinary differential equation:

$$h''' + hh'' = (h')^2. \tag{1.13}$$

The following is our main result on the classification of all solutions of equation (1.13):

Theorem 1.1. *Let h be any solution to equation (1.13) with prescribed initial data $h(0)$, $h'(0)$ and $h''(0)$. Then $h(x)$ belongs to one of the following types:*

(I) *$h(x)$ is a constant function on $\mathbb{R}$.*
(II) *$h'(x) > 0$ for all $x \in \mathbb{R}$, there exists some $r \in \mathbb{R}$ such that $h''(x) > 0 = h''(r)$ for all $x > r$, $h''(x) < 0$ for all $x < r$, $h(x)$ approaches to infinity at an exponential rate as x goes to infinity, and $h(x)$ approaches to negative infinity at an exponential rate as x goes to negative infinity.*
(III) *there are some constants $c_1 < r < c_2$ such that $h'(x) < 0 = h'(c_1) = h'(c_2)$ for all $c_1 < x < c_2$, $h'(x) > 0$ for all $x < c_1$ or $x > c_2$, $h''(x) > 0 = h''(r)$ for all $x > r$, $h''(x) < 0$ for all $x < r$, $h(x)$ approaches to infinity at an exponential rate as x goes to infinity, and $h(x)$ approaches to negative infinity at an exponential rate as x goes to negative infinity.*

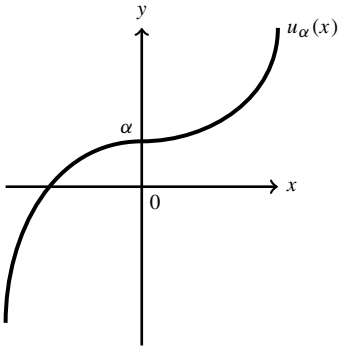


Fig. 1. $u_\alpha(x)$.

(IV) *there are some constants* a, $\alpha \in \mathbb{R}\backslash\{0\}$ *such that* $h(x) = ae^{\alpha x} - \alpha$ *for all* $x \in \mathbb{R}$.

(V) *there exists some* $0 < T < \infty$ *such that the maximal solution interval of* $h(x)$ *is* $(-\infty, T)$, $h^{(k)}(x) < 0$ *for all* $x < T$, *for each* $k = 0, 1, 2, 3, 4$, $\lim_{x \longrightarrow -\infty} h(x) = 0$, *and* $\lim_{x \longrightarrow T^-} h(x) = -\infty$.

(VI) $h'(x) < 0$ *and* $h''(x) < 0$ *for all* $x \in \mathbb{R}$, $\lim_{x \longrightarrow -\infty} h(x) = 0$, *and* $h(x)$ *approaches to negative infinity faster than an exponential rate as* x *goes to infinity.*

(VII) *there exists some* $0 < T < \infty$ *such that the maximal solution interval of* $h(x)$ *is* $(-T, \infty)$, $(-1)^{k+1}h^{(k)}(x) < 0$ *for all* $x > -T$, *for each* $k = 0, 1, 2, 3, 4$, $\lim_{x \longrightarrow \infty} h(x) = 0$, *and* $\lim_{x \longrightarrow -T^+} h(x) = \infty$.

(VIII) $h'(x) < 0$ *and* $h''(x) > 0$ *for all* $x \in \mathbb{R}$, $\lim_{x \longrightarrow \infty} h(x) = 0$, *and* $h(x)$ *approaches to infinity faster than an exponential rate as* x *goes to negative infinity.*

Let's give some interesting remarks about our main result.

Remark 1.1. For any $\alpha \in \mathbb{R}$, let u_α be the solution of the equation $h''' + hh'' = (h')^2$ with the initial data $h(0) = \alpha$, $h'(0) = 1$ and $h''(0) = 0$. By **Theorem 1.1**, we know that u_α must be the Type II solution (Fig. 1).

(1) Let $H(x) := -u_\alpha(-x)$, $x \geq 0$, it's easy to see that H satisfies $H''' + HH'' = (H')^2$, $H(0) = -\alpha$, $H'(0) = 1$ and $H''(0) = 0$, by the theorem of uniqueness for solutions to the ODE, we know that $-u_\alpha(-x) = H(x) = u_{-\alpha}(x)$ for all $x \in \mathbb{R}$. In particular, $u_0(x)$ is an odd function on $\mathbb{R}$.

(2) Let $h(x)$ be any Type II in **Theorem 1.1**, then $h'(x) > 0$ for all $x \in \mathbb{R}$ and there exists some $r \in \mathbb{R}$ such that $h''(r) = 0$. Let $\lambda = \frac{1}{\sqrt{h'(r)}}$ and $H(x) = \lambda h(\lambda x + r)$, $x \geq 0$, it's easy to see that H satisfies $H''' + HH'' = (H')^2$, $H(0) = \lambda h(r)$, $H'(0) = 1$ and $H''(0) = 0$, by the theorem of uniqueness for solutions to the ODE, we know that $\lambda h(\lambda x + r) = H(x) = u_{\lambda h(r)}(x)$ for all $x \in \mathbb{R}$, that is, $h(x) = \sqrt{h'(r)} \cdot u_{\frac{h(r)}{\sqrt{h'(r)}}}(\sqrt{h'(r)}(x - r))$, $x \in \mathbb{R}$.

Remark 1.2. For any $\alpha \in \mathbb{R}$, let v_α be the solution of the equation $h''' + hh'' = (h')^2$ with the initial data $h(0) = \alpha$, $h'(0) = -1$ and $h''(0) = 0$. By **Theorem 1.1**, we know that v_α must be the Type III solution (Fig. 2).

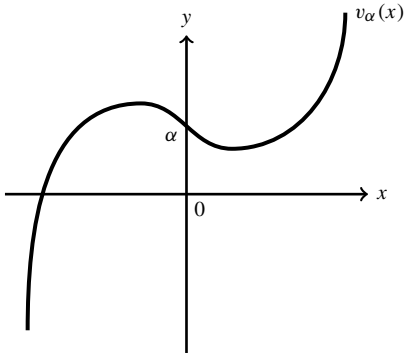
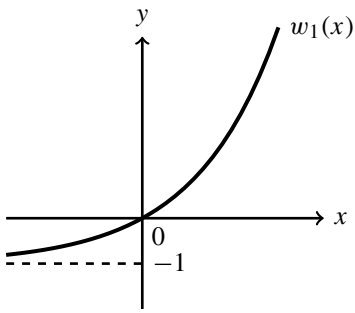


Fig. 2. $v_\alpha(x)$.

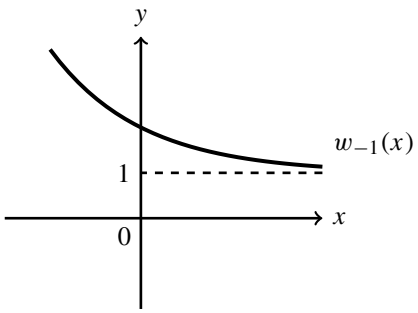
By a similar argument with **Remark 1.1**, we can obtain that

(1) $v_{-\alpha}(x) = -v_\alpha(-x)$ for all $x \in \mathbb{R}$. In particular, $v_0(x)$ is an odd function on $\mathbb{R}$.
(2) for any Type II solution $h(x)$ in **Theorem 1.1**, then there exists some $r \in \mathbb{R}$ such that $h(x) = \sqrt{-h'(r)} \cdot v_{\frac{h(r)}{\sqrt{-h'(r)}}}(\sqrt{-h'(r)}(x-r))$, $x \in \mathbb{R}$.

Remark 1.3. Let $w_1(x) = e^x - 1$ and $w_{-1}(x) = e^{-x} + 1$ for all $x \in \mathbb{R}$.



(a) The function $w_1(x) = e^x - 1$

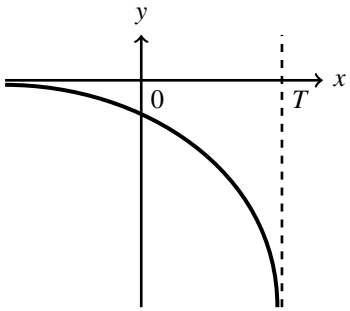


(b) The function $w_{-1}(x) = e^{-x} + 1$

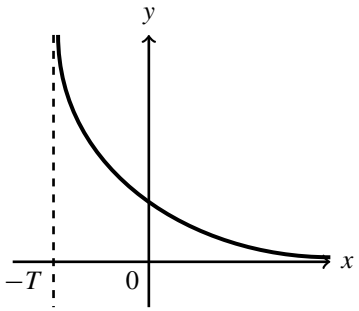
For any $a, \alpha \in \mathbb{R}\backslash\{0\}$, let $h_{a,\alpha}(x) = ae^{\alpha x} - \alpha$ for all $x \in \mathbb{R}$.

(1) If $a\alpha > 0$, then $h_{a,\alpha}(x) = \alpha w_1(\alpha x + \ln \frac{a}{\alpha})$ for all $x \in \mathbb{R}$.
(2) If $a\alpha < 0$, then $h_{a,\alpha}(x) = -\alpha w_{-1}(-\alpha x - \ln \frac{a}{-\alpha})$ for all $x \in \mathbb{R}$.

Remark 1.4. The function $h(x)$ is a Type V solution in **Theorem 1.1**, if and only if, $-h(-x)$ is a Type VII solution. Moreover, if **Conjecture 3.1** were true, by using the theorem of uniqueness for solutions to the ODE and the analysis in **Section 3** for bounded solutions on the right and left half lines, and **Theorem 1.1**, we know that there are no type VI solutions and type VIII solutions, only Type V solutions are of the form $h(x) = \frac{6}{x-T}$, and only Type VII solutions are of the form $h(x) = \frac{6}{x+T}$, where $T > 0$.



(a) The function $h(x) = \frac{6}{x-T}$ for $x < T$



(b) The function $h(x) = \frac{6}{x+T}$ for $x > -T$

In the following let's assume that $h(x)$ is a non-constant solution of equation (1), for each non-negative integer k, let's denote the k-th derivative of $h(x)$ by $h_k(x) := h^{(k)}(x)$. Hence the equation (1) can be rewritten as:

$$h_1'' + (h+\beta)h_1' - h'h_1 = 0. \tag{2}$$

Differentiating the above equation yields $h_1''' + h'h_1' + (h+\beta)h_1'' - h''h_1 - h'h_1' = 0$, that is,

$$h_2'' + (h+\beta)h_2' - h'h_2 = 0. \tag{3}$$

Hence both h_1 and h_2 satisfy the same homogeneous linear equation $y'' + (h+\beta)y' - h'y = 0$. This fact will play a very important role in our analysis later.

The following simple yet fundamental comparison principle will be used extensively in the subsequent analysis.

Lemma 1. *Let $c(x)$ be a continuous function on the interval $[a,b]$, and $y(x)$ satisfy the differential inequality $y' + c(x)y \geq 0$ for $a < x \leq b$.*

(a) *If $y(a) \geq 0$, then we have $y(x) \geq 0$ for all $x \in [a,b]$.*
(b) *If $y(a) > 0$, then we have $y(x) > 0$ for all $x \in [a,b]$.*
(c) *If $y' + c(x)y > 0$ for all $a < x \leq b$, and $y(a) \geq 0$, then we have $y(x) > 0$ for all $x \in (a,b]$.*

Proof. Since $y' + c(x)y \geq 0 (> 0)$, $x \in [a,b]$, by multiplying the integrating factor $e^{\int_0^x c(t)\,dt}$, we can obtain $\left(e^{\int_a^x c(t)\,dt} y\right)' \geq 0 (> 0)$, $x \in (a,b]$, which implies that the function $e^{\int_a^x c(t)\,dt} y(x)$ is non-decreasing(strictly increasing) on the interval $[a,b]$. Hence the statements follow straightforwardly by the monotonicity. □

This paper is organized as follows. In **Section 2**, we study the maximal interval of existence for solutions to equation (1) under various initial conditions for $h(0)$, $h'(0)$ and $h''(0)$. **Section 3** classifies the bounded solutions of the equation on the right half line and on the whole line. The properties of unbounded solutions are analyzed in **Section 4**. Finally, **Section 5** presents the proof of the main results, providing a complete classification of all bounded and unbounded solutions of (1).

2. Maximal solution intervals

In this section we study the maximal solution interval of the equation (1) under various initial data $h(0)$, $h'(0)$ and $h''(0)$. Ultimately, we will see that the maximal interval coincides with $\mathbb{R}$, except possibly when $h'(0) < 0$ and $h''(0) < 0$. Moreover, under the case that $h'(0) < 0$ and $h''(0) < 0$, we establish several fundamental properties of the solution.

By considering the function $h(x) + \beta$, W.L.O.G., we can assume that $\beta = 0$, that is, we will consider the following nonlinear ordinary differential equations:

$$h''' + hh'' = (h')^2. \tag{2.1}$$

To begin, consider the scenario where $h'''(x) \equiv 0$ on some open subinterval (a, b), that is, there are some constants $A, B, C \in \mathbb{R}$ such that $h(x) = Ax^2 + Bx + C$, $x \in (a, b)$, by plugging it into the equation (2.1), we obtain $0 + (Ax^2 + Bx + C) \cdot 2A = (2Ax + B)^2$, $x \in (a, b)$, that is, $2A^2x^2 + 2ABx + B^2 - 2AC = 0$, $x \in (a, b)$, which implies that $A = B = 0$. Hence $h(x) \equiv C$ is a constant function on (a, b). By the theorem of uniqueness for solutions to the ODE, we must have $h(x) \equiv C$ for all $x \in \mathbb{R}$.

2.1. Rightwards infinity solution intervals when $h'(0) \geq 0$ and $h''(0) \geq 0$

In this subsection, we consider a non-constant solution h of equation (2.1) satisfying $h'(0) \geq 0$ and $h''(0) \geq 0$, and its maximal forward interval of existence is taken to be $[0, T)$, where $0 < T \leq \infty$. We are going to show that $T = \infty$.

Proposition 2.1. *We have $h''(x) > 0$ for all $x \in (0, T)$. Consequently, given that $h'(0) \geq 0$, it follows that $h'(x) > 0$ for all $x \in (0, T)$.*

Proof. The equation (2.1) states that $h_2' + hh_2 = (h')^2 \geq 0$ for all $0 \leq x < T$, given that $h_2(0) = h''(0) \geq 0$, **Lemma 1** implies that $h''(x) = h_2(x) \geq 0$ for all $0 \leq x < T$. We now apply the strong maximum principle to equation (3) (with $\beta = 0$), which implies that

$$\text{either} \qquad h''(x) \equiv 0 \text{ for all } 0 \leq x < T, \qquad \text{or} \qquad h''(x) > 0 \text{ for all } 0 < x < T.$$

Since only linear solutions of (2.1) are constant functions, we conclude that

$$h''(x) > 0 \qquad \text{for all } 0 < x < T. \quad \square$$

Theorem 2.1. $T = \infty$.

Proof. We proceed by contradiction. If $0 < T < \infty$, then **Proposition 2.1** gives that $h'(x) > 0$ for all $0 < x < T$. Since $[0, T)$ is the rightwards maximal solution interval, it follows that $\lim_{x \longrightarrow T^-} h(x) = \infty$. Translating the solution if necessary, W.L.O.G., we can assume that

$$h_k(x) > 0 \quad \text{for all } 0 \leq x < T, \qquad \forall k = 0, 1, 2.$$

Define the function $g(x) := -\frac{h''(x)}{h'(x)}$, $0 \le x < T$, it is then straightforward to verify that

$$g(x) < 0, \quad g' + hg + h' - g^2 = 0, \quad \text{and} \quad g'' + (h - 2g)g' = 0 \qquad \text{for all } 0 \le x < T.$$

Since $g'(x)$ satisfies $y' + (h - 2g)y = 0$ on $[0, T)$, the theorem of uniqueness for solutions to the ODE implies that

$$\begin{aligned} &\text{either} \quad g'(x) \equiv 0 \text{ for all } 0 \le x < T, \quad \text{or} \quad g'(x) > 0 \text{ for all } 0 \le x < T, \\ &\quad \text{or} \quad g'(x) < 0 \text{ for all } 0 \le x < T. \end{aligned}$$

(a) If $g'(x) \equiv 0$ for all $0 \le x < T$. Then there exists some constant $\alpha > 0$ such that

$$h''(x) - \alpha h'(x) = 0 \qquad \text{for all } 0 \le x < T.$$

Hence there are some $a, b \in \mathbb{R}$ such that $h(x) = ae^{\alpha x} + b$ for all $0 \le x < T$. By the theorem of uniqueness for solutions to the ODE, we have $T = \infty$, contradiction.
(b) If $g'(x) < 0$ for all $0 \le x < T$. Since $g'' = -(h - 2g)g'$, $h(x) > 0$, $g(x) < 0$ and $g'(x) < 0$ for all $0 \le x < T$, then

$$g''(x) > 0 \qquad \text{for all } 0 \le x < T.$$

Since $g'(x) < 0$ for all $0 \le x < T$, then there exists some $B \le 0$ such that

$$\lim_{x \longrightarrow T^-} g'(x) = B, \qquad \text{and} \qquad g'(x) < B \quad \text{for all } 0 \le x < T.$$

By the mean value theorem, there exists an increasing sequence $\{x_n\}_{n=1}^{\infty} \subseteq (0, T)$ such that

$$\lim_{n \longrightarrow \infty} x_n = T, \qquad \text{and} \qquad \lim_{n \longrightarrow \infty} g''(x_n) = 0.$$

Since $g'' = -(h - 2g)g'$ for all $0 \le x < T$, we have

$$\lim_{n \longrightarrow \infty} (h(x_n) - 2g(x_n))g'(x_n) = 0.$$

Since $\lim_{n \longrightarrow \infty} x_n = T$, $\lim_{x \longrightarrow T^-} h(x) = \infty$, $\lim_{x \longrightarrow T^-} g'(x) = B$, and $g(x) < 0$ for all $0 \le x < T$, then

$$\lim_{x \longrightarrow T^-} g'(x) = B = 0.$$

Since $g'' = -(h - 2g)g'$ for all $0 \le x < T$, $\lim_{x \longrightarrow T^-} h(x) = \infty$ and $g(x) < 0$ for all $0 \le x < T$, we have

$$\lim_{x \longrightarrow T^-} \frac{g''(x)}{g'(x)} = \lim_{x \longrightarrow T^-} -(h(x) - 2g(x)) = -\infty.$$

Since $g'(x) < 0$ for all $0 \le x < T$, then there exists some $\alpha \in (0, T)$ such that

$$(\ln(-g'(x)))' = \frac{g''(x)}{g'(x)} \le -1 \qquad \text{for all } \alpha \le x < T.$$

Hence $\ln \frac{g'(x)}{g'(\alpha)} \le -(x-\alpha)$ for all $\alpha \le x < T$. Since $g'(x) < 0$ for all $0 \le x < T$, then

$$g'(x) \ge g'(\alpha) e^{-x+\alpha} \qquad \text{for all } \alpha \le x < T.$$

Integrating the above inequality yields that $g(x) \ge -g'(\alpha)(e^{-x+\alpha} - 1) - g(\alpha)$ for all $\alpha \le x < T$. Since $0 < T < \infty$, then $g(x)$ is bounded from below on $[0, T)$. Since $g(x) = -\frac{h''(x)}{h'(x)}$ for all $0 \le x < T$, then $\frac{h''(x)}{h'(x)}$ is bounded from above on $[0, T)$, that is, there exists some $\beta > 0$ such that

$$(\ln(h'(x)))' = \frac{h''(x)}{h'(x)} \le \beta \qquad \text{for all } 0 \le x < T.$$

Hence $\ln \frac{h'(x)}{h'(0)} \le \beta x$ for all $0 \le x < T$. Since $h'(x) > 0$ for all $0 \le x < T$, then $h'(x) \le h'(0)e^{\beta x}$ for all $0 \le x < T$. Integrating this inequality we obtain $h(x) \le \frac{h'(0)}{\beta}(e^{\beta x} - 1) + h(0)$ for all $0 \le x < T$. Since $0 < T < \infty$, then $h(x)$ is bounded from above on $[0, T)$, which contradicts with the fact that $\lim_{x \longrightarrow T^-} h(x) = \infty$.

(c) If $g'(x) > 0$ for all $0 \le x < T$. Since $g(x) < 0$ for all $0 \le x < T$, it's easy to see that $g(x)$ is bounded on $[0, T)$. Following the same reasoning used to conclude the proof of case (b), we arrive at a contradiction.

In summary we can conclude that $T = \infty$. □

2.2. *Rightwards infinity solution intervals when $h'(0) < 0$ and $h''(0) \ge 0$*

In this subsection let's assume that h is a non-constant solution of equation (2.1) with the conditions $h'(0) < 0$ and $h''(0) \ge 0$, and the rightwards maximal solution interval is $[0, T)$ with $0 < T \le \infty$. We are going to show that $T = \infty$.

Proposition 2.2. *We have $h''(x) > 0$ for all $0 < x < T$.*

Proof. The proof is the same as **Proposition 2.1**. □

Theorem 2.2. $T = \infty$.

Proof. If the statement is not true, we have $0 < T < \infty$. By **Proposition 2.2**, we have $h''(x) > 0$ for all $0 < x < T$. Apply the result in **Subsection 2.1** to a right shift of $h(x)$, we only need to show that there exists some $x_0 \in (0, T)$ such that

$$h'(x_0) = 0.$$

If this is not true, since $h'(0) < 0$, by the continuity, we must have $h'(x) < 0$ for all $0 \le x < T$. Since $h''(x) > 0$ for all $0 < x < T$, it's easy to know that $h'(x)$ is a bounded function on $[0, T)$.

Since $0 < T < \infty$, we know that $h(x)$ is a bounded function on $[0, T)$. On the other hand, since $h'(x) < 0$ for all $0 < x < T$ and $[0, T)$ is the right maximal solution interval, we should have $\lim_{x \longrightarrow T^-} h(x) = -\infty$, contradiction. □

2.3. Possible rightwards finite solution interval when $h'(0) \leq 0$ and $h''(0) < 0$

In this subsection let's assume that h is a non-constant solution of equation (2.1) with the conditions $h'(0) \leq 0$ and $h''(0) < 0$, and the rightwards maximal solution interval is $[0, T)$ with $0 < T \leq \infty$.

Since $h'(0) \leq 0$ and $h''(0) < 0$, we know that

$$\text{either } h'(x) < 0 \text{ for all } 0 < x < T, \text{ or there exists some } 0 < c < T$$
$$\text{such that } h'(x) < 0 = h'(c) \text{ for all } 0 < x < c.$$

For the first case that $h'(x) < 0$ for all $0 < x < T$, it's possible to have a finite solution interval, for example, $h(x) = \frac{6}{x-1}$ for all $0 \leq x < 1$. For the second case that there exists some $0 < c < T$ such that $h'(x) < 0 = h'(c)$ for all $0 < x < c$, we are going to show that $T = \infty$.

Theorem 2.3. *If there exists some $0 < c < T$ such that $h'(x) < 0 = h'(c)$ for all $0 < x < c$, we have $T = \infty$.*

Proof. Since $h'(x) < 0 = h'(c)$ for all $0 < x < c$, we have $h''(c) \geq 0$. Since $h'(c) = 0$ and $h''(c) \geq 0$, apply the result in **Subsection 2.1** to the solution $h(x + c)$, we get $T = \infty$. □

Proposition 2.3. *If $0 < T < \infty$, then $h'(x) < 0$ for all $0 < x < T$, and $\lim_{x \longrightarrow T^-} h(x) = -\infty$.*

Proof. The statement follows directly from **Theorem 2.3**. □

Proposition 2.4. *If $0 < T < \infty$, then $h''(x) < 0$ for all $0 \leq x < T$.*

Proof. If not, since $h''(0) < 0$, then there exists some $0 < a < T$ such that $h''(a) = 0$. Apply the result in **Subsection 2.2** to the solution $h(x + a)$, we get $T = \infty$, contradiction. □

Proposition 2.5. *If $0 < T < \infty$, then $\lim_{x \longrightarrow T^-} h'(x) = -\infty$.*

Proof. Since $h'(x) < 0$ and $h''(x) < 0$ for all $0 \leq x < T$, then there exists some $A \in [-\infty, 0)$ such that $\lim_{x \longrightarrow T^-} h'(x) = A$. If the statement is not true, that is, $-\infty < A < 0$. Since $h''(x) < 0$ and $h'(x) < 0$ for all $0 \leq x < T$, it's easy to see that $h'(x)$ is bounded on $[0, T)$. Since $0 < T < \infty$, by integrating, we obtain that $h(x)$ is bounded on $[0, T)$, which contradicts with **Proposition 2.3**. □

Proposition 2.6. *If $0 < T < \infty$, then $h'''(x) < 0$ for all $0 \leq x < T$.*

Proof. If not, that is, there exists some $0 \leq a < T$ such that $h_3(a) = h'''(a) \geq 0$. Since $h'(x) < 0$ and $h''(x) < 0$ for all $0 < x < T$, we have $h_3' + hh_3 = h'h'' > 0$ for all $0 < x < T$. By **Lemma 1**,

we know that $h'''(x) > 0$ for all $a < x < T$. Since $h''(x) < 0$ for all $0 \le x < T$, it's easy to know that $h''(x)$ is bounded on $[0, T)$. Since $0 < T < \infty$, we know that $h(x)$ is bounded on $[0, T)$, which contradicts with the fact that $\lim_{x \longrightarrow T^-} h(x) = -\infty$. □

Proposition 2.7. *If* $0 < T < \infty$, *then* $\lim_{x \longrightarrow T^-} h''(x) = -\infty$.

Proof. The proof is similar to **Proposition 2.5**. □

Proposition 2.8. *If* $0 < T < \infty$, *then* $h_4(x) < 0$ *for all* $0 \le x < T$.

Proof. If not, that is, there exists some $0 \le a < T$ such that $h_4(a) \ge 0$. Since $h''(x) < 0$ for all $0 \le x < T$, we have $h_4' + hh_4 = (h'')^2 > 0$ for all $0 \le x < T$. By **Lemma 1**, we know that $h_4(x) > 0$ for all $a < x < T$. Since $h'''(x) < 0$ for all $0 \le x < T$, it's easy to know that $h'''(x)$ is bounded on $[0, T)$. Since $0 < T < \infty$, we know that $h(x)$ is bounded on $[0, T)$, which contradicts with the fact that $\lim_{x \longrightarrow T^-} h(x) = -\infty$. □

In the following of this subsection, let's assume that $0 < T < \infty$, hence we have $h_k(x) < 0$ for all $0 < x < T$, for each $k = 1, 2, 3, 4$. Let's define the function $g(x) := -\frac{h''(x)}{h'(x)}$, $0 < x < T$, so we obtain

$$g(x) < 0, \qquad g' + hg + h' - g^2 = 0, \quad \text{and} \quad g'' + (h - 2g)g' = 0 \qquad \text{for all } 0 < x < T.$$

Since the derivative $g'(x)$ solves $y' + (h - 2g)y = 0$, $0 < x < T$, by the theorem of uniqueness for solutions to the ODE, we have

$$\begin{aligned} &\text{either} \quad g'(x) \equiv 0 \text{ for all } 0 < x < T, \quad \text{or} \quad g'(x) > 0 \text{ for all } 0 < x < T, \\ &\text{or} \quad g'(x) < 0 \text{ for all } 0 < x < T. \end{aligned} \tag{2.2}$$

Proposition 2.9. *There holds that* $g'(x) < 0$ *for all* $0 < x < T$.

Proof. If not, by the alternative in (2.2), we must have $g'(x) \ge 0$ for all $0 < x < T$. Take $0 < \alpha < T$ and fix it, since $h'(x) < 0$ and $g(x) < 0$ for all $0 < x < T$, then $(\ln(-h'(x)))' = \frac{h''(x)}{h'(x)} = -g(x)$ is bounded on $[\alpha, T)$, that is, there exists some $\beta > 0$ such that

$$(\ln(-h'(x)))' = \frac{h''(x)}{h'(x)} \le \beta \qquad \text{for all } \alpha \le x < T.$$

Hence $\ln \frac{h'(x)}{h'(\alpha)} \le \beta(x - \alpha)$ for all $\alpha \le x < T$. Since $h'(x) < 0$ for all $0 \le x < T$, then $h'(x) \ge h'(\alpha)e^{\beta(x-\alpha)}$ for all $\alpha \le x < T$, which implies that $h(x) \ge \frac{h'(\alpha)}{\beta}(e^{\beta(x-\alpha)} - 1) + h(\alpha)$ for all $\alpha \le x < T$. Since $0 < T < \infty$, then $h(x)$ is bounded from below on $[0, T)$, which contradicts with the fact that $\lim_{x \longrightarrow T^-} h(x) = -\infty$. □

Proposition 2.10. *There holds that* $\lim_{x \longrightarrow T^-} g(x) = -\infty$.

Proof. By **Proposition 2.9** we have $g'(x) < 0$ for all $0 < x < T$. Since $g(x) < 0$ for all $0 < x < T$, then there exists some $A \in [-\infty, 0)$ such that $\lim_{x \longrightarrow T^-} g(x) = A$. If the statement is not true, that is, $-\infty < A < 0$. Take $0 < \alpha < T$ and fix it, since $g'(x) < 0$ and $g(x) < 0$ for all $0 < x < T$, then $-g(x)$ is bounded on $[\alpha, T)$. Since $0 < T < \infty$, by the same argument as **Proposition 2.9**, we will get a contradiction. □

Proposition 2.11. *There holds that $h'(0) < 0$. In particular, we have $h'(x) < 0$ for all $0 \le x < T$.*

Proof. If not, since $h'(0) \le 0$, we have $h'(0) = 0$. Since $h''(0) < 0$, $h'(x) < 0$ and $g(x) = -\frac{h''(x)}{h'(x)}$ for all $0 < x < T$, we obtain $\lim_{x \longrightarrow 0^+} g(x) = -\infty$. Since $\lim_{x \longrightarrow T^-} g(x) = -\infty$, by the mean value theorem, there exists some $x_0 \in (0, T)$ such that $g'(x_0) = 0$, which contradicts with the fact that $g'(x) < 0$ for all $0 < x < T$. □

By **Proposition 2.11**, we know that $g(x)$ is well-defined on $[0, T)$. Hence we have

$$g(x) < 0 \quad \text{and} \quad g'(x) < 0 \qquad \text{for all } 0 \le x < T.$$

Proposition 2.12. *There holds that $h(x) < 0$ for all $0 \le x < T$.*

Proof. Since $h'(x) < 0$ for all $0 \le x < T$, we can define $f(x) := \frac{h(x)}{h'(x)}$ for all $0 \le x < T$, since $h'''(x) < 0$ for all $0 \le x < T$, then

$$f'(x) = \frac{(h'(x))^2 - h(x)h''(x)}{(h'(x))^2} = \frac{h'''(x)}{(h'(x))^2} < 0 \qquad \text{for all } 0 \le x < T.$$

Since $\lim_{x \longrightarrow T^-} h(x) = \lim_{x \longrightarrow T^-} h'(x) = -\infty$, by L'Hospital's rule, then

$$0 = \lim_{x \longrightarrow T^-} -\frac{1}{g(x)} = \lim_{x \longrightarrow T^-} \frac{h'(x)}{h''(x)} = \lim_{x \longrightarrow T^-} \frac{h(x)}{h'(x)} = \lim_{x \longrightarrow T^-} f(x).$$

Since $f'(x) < 0$ for all $0 \le x < T$, then $f(x) > 0$ for all $0 \le x < T$. Since $f(x) = \frac{h(x)}{h'(x)}$ and $h'(x) < 0$ for all $0 \le x < T$, then

$$h(x) < 0 \qquad \text{for all } 0 \le x < T. \quad \square$$

Theorem 2.4. *The whole maximal solution interval is $(-\infty, T)$. Moreover, we have $h_k(x) < 0$ for all $-\infty < x < T$, for each $k = 0, 1, 2, 3, 4$.*

Proof. Assume that the left maximal interval of existence is $(\widetilde{T}, 0]$ for some $\widetilde{T} \in [-\infty, 0)$, let $H(x) := -h(-x)$, $0 \le x < -\widetilde{T}$, it's easy to know that $H'(0) = h'(0) \le 0$, $H''(0) = -h''(0) > 0$, and $H''' + HH'' = (H')^2$ for all $0 \le x < -\widetilde{T}$. By the results in **Subsection 2.1** and **Subsection 2.2**, we know that $-\widetilde{T} = \infty$, that is, $\widetilde{T} = -\infty$. Hence

the whole maximal interval of existence is $(-\infty, T)$.

By a left shifting consideration, by the previous propositions, we know that

$$h^{(k)}(x) < 0 \text{ for all } -\infty < x < T, \text{ for each } k = 0, 1, 2, 3, 4. \quad \square$$

2.4. Rightwards infinity solution intervals when $h'(0) > 0$ and $h''(0) < 0$

In this subsection let's assume that h is a non-constant solution of equation (2.1) with the conditions $h'(0) > 0$ and $h''(0) < 0$, and the rightwards maximal solution interval is $[0, T)$ with $0 < T \leq \infty$. We are going to show that $T = \infty$.

Since $h'(0) > 0$ and $h''(0) < 0$, we know that

$$\text{either } h'(x) > 0 \text{ for all } 0 \leq x < T, \text{ or there exists some } 0 < c < T$$
$$\text{such that } h'(x) > 0 = h'(c) \text{ for all } 0 \leq x < c.$$

Theorem 2.5. *There holds that $T = \infty$.*

Proof. Let's prove this in three cases. (a) If there exists some $0 < a < T$ such that $h''(a) \geq 0$. Apply the results in **Subsection 2.1** and **Subsection 2.2** to the solution $h(x + a)$, we know that $T = \infty$.

(b) If $h''(x) < 0$ and $h'(x) > 0$ for all $0 \leq x < T$. Hence $h'(x)$ is bounded on $[0, T)$. If $0 < T < \infty$, then $h(x)$ is bounded on $[0, T)$. On the other hand, since $h'(x) > 0$ for all $0 \leq x < T$, and $[0, T)$ is the rightwards maximal solution interval, we must have $\lim_{x \longrightarrow T^-} h(x) = \infty$, contradiction.

(c) If $h''(x) < 0$ for all $0 \leq x < T$, and there exists some $0 < c < T$ such that $h'(x) > 0 = h'(c)$ for all $0 \leq x < c$, which implies that $h''(c) < 0$. Since $h'(c) = 0$ and $h''(c) < 0$, apply the result in **Subsection 2.3** to the solution $h(x + c)$, we know that $T = \infty$.

In summary we can conclude that $T = \infty$. $\square$

2.5. Maximal solution intervals

To study the leftwards maximal solution interval of equation (2.1) under various initial data $h(0)$, $h'(0)$ and $h''(0)$, notice that the function $H(x) := -h(-x)$ also satisfies the equation (2.1) provided $h(x)$ is a solution of equation (2.1), so when applying the results in **Subsections 2.1, 2.2, 2.3** and **2.4** to the function $H(x)$, we can obtain the information of the leftwards maximal solution interval $(-T, 0]$ of equation (2.1).

Theorem 2.6. *Let h be any solution to equation (2.1) with given initial data $h(0)$, $h'(0)$ and $h''(0)$.*

(a) *If $h'(0) \geq 0$ and $h''(0) \leq 0$, then the leftwards maximal solution interval is $(-\infty, 0]$.*
(b) *If $h'(0) < 0$ and $h''(0) \leq 0$, then the leftwards maximal solution interval is $(-\infty, 0]$.*
(c) *If $h'(0) \leq 0$ and $h''(0) > 0$, then the leftwards maximal solution interval is either $(-\infty, 0]$ or $(-T, 0]$ for some $0 < T < \infty$. In the case that $0 < T < \infty$, we must have $(-1)^{k+1} h_k(x) < 0$ for all $-T < x \leq 0$, for each $k = 0, 1, 2, 3, 4$, and $\lim_{x \longrightarrow -T^+} h(x) = \infty$.*
(d) *If $h'(0) > 0$ and $h''(0) > 0$, then the leftwards maximal solution interval is $(-\infty, 0]$.*

Combining the results on rightwards and leftwards maximal solution intervals, we obtain the following theorem about the maximal solution intervals of solutions to equation (2.1).

Theorem 2.7. *Let h be any solution to equation (2.1) with given initial data $h(0)$, $h'(0)$ and $h''(0)$.*

(a) *If $h'(0) \geq 0$ and $h''(0) \leq 0$, then the maximal solution interval is $\mathbb{R}$.*
(b) *If $h'(0) < 0$ and $h''(0) \leq 0$, then the maximal solution interval is either $\mathbb{R}$ or $(-\infty, T)$ for some $0 < T < \infty$. In the case that $0 < T < \infty$, we must have $h^{(k)}(x) < 0$ for all $x < T$, for each $k = 0, 1, 2, 3, 4$.*
(c) *If $h'(0) \leq 0$ and $h''(0) > 0$, then the maximal solution interval is either $\mathbb{R}$ or $(-T, \infty)$ for some $0 < T < \infty$. In the case that $0 < T < \infty$, we must have $(-1)^{k+1}h^{(k)}(x) < 0$ for all $x > -T$, for each $k = 0, 1, 2, 3, 4$.*
(d) *If $h'(0) > 0$ and $h''(0) > 0$, then the maximal solution interval is $\mathbb{R}$.*

3. Bounded solutions

This section is devoted to the bounded solutions of equation (1). We will establish that the only bounded solutions existing on the whole real line are constants. Two observations allow us to simplify the analysis. First, if $h(x)$ solves (1), then $h(x) + \beta$ satisfies (2.1). Second, if $h(x)$ satisfies (2.1), then so does the function $H(x) := -h(-x)$. Hence it is sufficient to study bounded solutions on the right half line $[0, \infty)$, that is,

$$h''' + (h + \beta)h'' = (h')^2 \qquad \text{for all } x \geq 0. \tag{3.1}$$

3.1. Bounded solutions on the right half line when $h'(0) \leq 0$

In this subsection let's assume that $h(x)$ is a bounded non-constant solution of equation (3.1) with the condition $h'(0) \leq 0$. In this case, In this case, after normalizing $h(x)$ by adding a constant so that $\lim_{x \longrightarrow \infty} h(x) = 0$, we must have $\beta \geq 0$. Furthermore if $\beta > 0$, then $h(x) = ae^{-\beta x}$ for some constant $a > 0$.

Lemma 3.1. *There holds that $h'(x) < 0$ for all $x > 0$. In particular, since h is a bounded function on $[0, \infty)$, we know that the limit $h(\infty) := \lim_{x \longrightarrow \infty} h(x)$ exists.*

Proof. Since $h'(x)$ satisfies the equation (2) and $h(x)$ is a non-constant function on $[0, \infty)$, by the strong maximum principle or the theorem of uniqueness for solutions to the ODE, it suffices to show that

$$h'(x) \leq 0 \qquad \text{for all } x > 0.$$

If the above assertion is not true, that is, there exists some $a > 0$ such that $h'(a) > 0$. Since $h'(0) \leq 0$, we know that there is some $0 \leq b < a$ such that $h'(x) > 0 = h'(b)$ for all $b < x \leq a$, which implies that $h''(b) \geq 0$. Since $h_1(b) = 0$, h_1 satisfies the equation (2), and h is not a constant function on $[0, \infty)$, by the theorem of uniqueness for solutions to the ODE we must have

$$h''(b) > 0.$$

On the other hand, since $h_2' + (h+\beta)h_2 = (h')^2 \geq 0$, $x \geq 0$, **Lemma 1** tells us that $h''(x) > 0$ for all $x \geq b$. Since $a > b$, we have $h'(x) \geq h'(a) > 0$ for all $x \geq a$, which implies that $h(x) \geq h'(a)(x-a)+h(a)$ for all $x \geq a$. Since $h'(a) > 0$, we obtain $\lim_{x\longrightarrow\infty} h(x) = \infty$, which contradicts with the assumption that h is a bounded solution of equation (3.1). □

By considering $h(x) - h(\infty)$, W.L.O.G., in the following of this subsection let's assume that

$$h(\infty) = \lim_{x\longrightarrow\infty} h(x) = 0. \tag{3.2}$$

Lemma 3.2. *There exists some $R_1 > 0$ such that $h''(x) > 0$ for all $x \geq R_1$.*

Proof. The statement is established by splitting the argument into two cases. If there exists some $a > 0$ such that $h_2(a) = h''(a) > 0$, since $h_2' + (h+\beta)h_2 = (h')^2 \geq 0$, $x \geq 0$, by **Lemma 1**, we should have $h''(x) = h_2(x) > 0$ for all $x \geq a$. By taking $R_1 = a$, we are done.

If $h''(x) \leq 0$ for all $x \geq 0$, then $h'(x) \leq h'(1)$ for all $x \geq 1$. Integrating this inequality gives us $h(x) \leq h'(1)(x-1) + h(1)$ for all $x \geq 1$. Since $h'(1) < 0$, we can obtain $\lim_{x\longrightarrow\infty} h(x) = -\infty$, which contradicts with the limit (3.2). □

By **Lemma 3.2** and the limit (3.2), applying the mean value theorem, we can obtain

$$\lim_{x\longrightarrow\infty} h'(x) = 0. \tag{3.3}$$

Proposition 3.1. *There holds that $\lim_{x\longrightarrow\infty} h''(x) = 0$.*

Proof. Let's consider the function $\phi(x) := h''(x) + \beta h'(x) + h(x)h'(x)$, $x \geq 0$, by using the equation (3.1) it's easy to see that $\phi' = h''' + \beta h'' + (h')^2 + hh'' = h''' + (h+\beta)h'' + (h')^2 = 2(h')^2 \geq 0$. Hence

$$\phi(x) \text{ is non-decreasing on } [0, \infty).$$

Since $\lim_{x\longrightarrow\infty} h'(x) = 0$, by the mean value theorem, we can find an increasing sequence $\{x_n\}_{n=1}^{\infty} \subseteq [0, \infty)$ such that $\lim_{n\longrightarrow\infty} x_n = \infty$ and $\lim_{n\longrightarrow\infty} h''(x_n)=0$. Since $\lim_{x\longrightarrow\infty} h(x)= \lim_{x\longrightarrow\infty} h'(x) = 0$, by the definition of ϕ, we have $\lim_{n\longrightarrow\infty} \phi(x_n) = 0$. Since ϕ is non-decreasing on $[0, \infty)$ and $\lim_{n\longrightarrow\infty} x_n = \infty$, we obtain

$$\lim_{x\longrightarrow\infty} \phi(x) = 0.$$

Since $\lim_{x\longrightarrow\infty} h(x) = \lim_{x\longrightarrow\infty} h'(x) = 0$, by the definition of ϕ, we have

$$\lim_{x\longrightarrow\infty} h''(x) = 0. \quad \square$$

Proposition 3.2. *There holds that* $\lim_{x\longrightarrow\infty} h^{(k)}(x) = 0$ *for all* $k \in \mathbb{N}_0$.

Proof. The statements follow directly from **Proposition 3.1**, the lmits (3.2) and (3.3), the equation (3.1), and differentiating the equation (3.1). □

Proposition 3.3. *There holds that* $\beta \geq 0$.

Proof. If the statement is not true, that is, $\beta < 0$. By equation (3.1), we know that

$$h''' = (-h-\beta)h'' + (h')^2 \geq (-\beta - h)h'' \qquad \text{for all } x \geq 0.$$

Since $h''(x) > 0$ for all $x \geq R_1$, $\lim_{x\longrightarrow\infty} h(x) = 0$ and $\beta < 0$, then there exists some $R_2 > R_1$ such that $-\beta - h(x) \geq -\frac{\beta}{2}$ for all $x \geq R_2$, which implies that $h''' \geq -\frac{\beta}{2}h'' > 0$, $x \geq R_2$, which implies that $h''(x) \geq h''(R_2) > 0$ for all $x \geq R_2$. Since $h'''(x) > 0$ for all $x \geq R_2$, we have $\liminf_{x\longrightarrow\infty} h''(x) \geq h''(R_2) > 0$, which contradicts with the limit $\lim_{x\longrightarrow\infty} h''(x) = 0$. □

Since $h'(x) < 0$ for all $x > 0$, and both $h'(x)$ and $h''(x)$ satisfy the same homogeneous second order linear equation $y'' + (h+\beta)y' - h'y = 0$, it's natural to consider the function $g(x) := -\frac{h''(x)}{h'(x)}$, $x > 0$. It's easy to see that

$$g' + (h+\beta)g + h' - g^2 = 0, \quad \text{and} \quad g'' + (h+\beta-2g)g' = 0 \qquad \text{for all } x > 0. \tag{3.4}$$

Since g' is a solution of the first order linear differential equation $y' + (h+\beta-2g)y = 0$, $x > 0$, applying the theorem of uniqueness for solutions to the ODE, we have the following alternative:

$$\text{either} \qquad g'(x) \equiv 0 \text{ for all } x > 0, \quad \text{or} \quad g'(x) < 0 \text{ for all } x > 0, \quad \text{or} \quad g'(x) > 0 \text{ for all } x > 0. \tag{3.5}$$

Lemma 3.3. *There holds that either* $g'(x) < 0$ *for all* $x > 0$, *or* $g'(x) \equiv 0$ *for all* $x > 0$.

Proof. If the assertion is not true, by the alternative in (3.5), we must have

$$g'(x) > 0 \qquad \text{for all } x > 0.$$

Since $h'(x) < 0$ for all $x > 0$, $h''(x) > 0$ for all $x \geq R_1$, and $g(x) = -\frac{h''(x)}{h'(x)}$, we know that $g(x) > 0$ for all $x \geq R_1$. Since $g'(x) > 0$ for all $x > 0$, we obtain

$$\text{the limit } \lim_{x\longrightarrow\infty} \frac{1}{g(x)} \text{ exists.} \tag{3.6}$$

By using the equation (3.1) and the definition of g, we have

$$-\frac{h'''}{h''} = \frac{(h+\beta)h'' - (h')^2}{h''} = h + \beta - h' \cdot \frac{h'}{h''} = h + \beta + \frac{h'}{g}.$$

Since $\lim_{x\longrightarrow\infty} h(x) = \lim_{x\longrightarrow\infty} h'(x) = 0$, by the limit (3.6), we have $\lim_{x\longrightarrow\infty} -\frac{h'''(x)}{h''(x)} = \beta$. Since $\lim_{x\longrightarrow\infty} h'(x) = \lim_{x\longrightarrow\infty} h''(x) = 0$, by L'Hospital's rule, we obtain

$$\lim_{x\longrightarrow\infty} g(x) = \lim_{x\longrightarrow\infty} -\frac{h''(x)}{h'(x)} = \lim_{x\longrightarrow\infty} -\frac{h'''(x)}{h''(x)} = \beta.$$

Since $g(R_1) > 0$, and $g'(x) > 0$ for all $x > 0$, hence we must have

$$\beta > 0.$$

Let's consider the function $c(x) := h(x) + \beta - 2g(x)$, $x > 0$, since $\lim_{x\longrightarrow\infty} h(x) = 0$ and $\lim_{x\longrightarrow\infty} g(x) = \beta$, we have $\lim_{x\longrightarrow\infty} c(x) = -\beta < 0$, which implies that there is some $R_2 > R_1$ such that $c(x) \le -\frac{\beta}{2}$ for all $x \ge R_2$. The second equation in (3.4) gives $g'(x) = e^{-\int_{R_2}^{x} c(t)\,dt} g'(R_2)$, $x \ge R_2$. Since $g'(x) > 0$ for all $x > 0$, and $c(x) \le -\frac{\beta}{2}$ for all $x \ge R_2$, we obtain $g'(x) \ge g'(R_2)e^{\int_{R_2}^{x} \frac{\beta}{2}\,dt} = g'(R_2)e^{\frac{\beta}{2}(x-R_2)}$ for all $x \ge R_2$. Since $\beta > 0$ and $g'(R_2) > 0$, it's easy to obtain $\lim_{x\longrightarrow\infty} g'(x) = \infty$, which contradicts with the limit $\lim_{x\longrightarrow\infty} g(x) = \beta$. □

Lemma 3.4. *There holds that $h''(x) > 0$ for all $x > 0$.*

Proof. If the statement is not true, that is, there is some $a > 0$ such that $h''(a) \le 0$. By **Lemma 3.2**, we know that $a < R_1$ and there exists some $R_2 \in [a, R_1)$ such that $h''(R_2) = 0$ and $h''(x) > 0$ for all $x > R_2$. Since $g(x) = -\frac{h''(x)}{h'(x)}$ and $h'(x) < 0$ for all $x > 0$, we have $g(R_2) = 0$ and $g(x) > 0$ for all $x > R_2$. Hence we obtain $g'(R_2) \ge 0$. By **Lemma 3.3**, we should have $g'(x) \equiv 0$ for all $x > 0$. Since $g(R_2) = 0$, we obtain $g(x) \equiv 0$ for all $x > 0$, contradiction. □

Since $\lim_{x\longrightarrow\infty} h(x) = \lim_{x\longrightarrow\infty} h'(x) = 0$, and $h''(x) > 0$ for all $x > 0$, we obtain

$$h'(x) < 0 \quad \text{and} \quad h(x) > 0 \qquad \text{for all } x \ge 0. \tag{3.7}$$

Hence the function $g(x) = -\frac{h''(x)}{h'(x)}$ is well-defined for all $x \ge 0$, and $g(x)$ satisfies

$$g' + (h+\beta)g + h' - g^2 = 0, \quad \text{and} \quad g'' + (h+\beta-2g)g' = 0 \qquad \text{for all } x \ge 0. \tag{3.8}$$

Since $h''(x) > 0$ and $h'(x) < 0$ for all $x > 0$, we have

$$g(x) > 0 \qquad \text{for all } x > 0. \tag{3.9}$$

Since g' is a solution of the first-order linear differential equation $y' + (h + \beta - 2g)y = 0$, $x \ge 0$, by the theorem of uniqueness for solutions to the ODE, and **Lemma 3.3**, we obtain

$$\text{either} \qquad g'(x) \equiv 0 \text{ for all } x \ge 0, \qquad \text{or} \qquad g'(x) < 0 \text{ for all } x \ge 0. \tag{3.10}$$

Proposition 3.4. *If $g'(x) \equiv 0$ for all $x \geq 0$, then we have $\beta > 0$ and $h(x) = ae^{-\beta x}$ for all $x \geq 0$, where $a := h(0) > 0$.*

Proof. Since $g'(x) \equiv 0$ for all $x \geq 0$, we know that $-\frac{h''(x)}{h'(x)} = g(x) = g(1) := \gamma > 0$, $x \geq 0$, that is, $h''(x) + \gamma h'(x) = 0$, $x \geq 0$. It's easy to know that there are some $a, b \in \mathbb{R}$ such that $h(x) = ae^{-\gamma x} + b$ for all $x \geq 0$. Since $\lim_{x \longrightarrow \infty} h(x) = 0$, $h(x) > 0$ for all $x \geq 0$, and $\gamma > 0$, we must have $a > 0$, $b = 0$ and $h(x) = ae^{-\gamma x}$ for all $x \geq 0$. Plugging $h(x) = ae^{-\gamma x}$ into the equation (3.1), we obtain $-a\gamma^3 e^{-\gamma x} + (ae^{-\gamma x} + \beta) \cdot a\gamma^2 e^{-\gamma x} = (-a\gamma e^{-\gamma x})^2$ for all $x \geq 0$, that is, $a\gamma^2 e^{-\gamma x}(-\gamma + \beta) = 0$ for all $x \geq 0$. Since $a > 0$ and $\gamma > 0$, we obtain

$$\beta = \gamma > 0, \qquad \text{and} \qquad h(x) = ae^{-\beta x} \quad \text{for all } x \geq 0. \quad \square$$

Lemma 3.5. *If $g'(x) < 0$ for all $x \geq 0$, there holds that $\lim_{x \longrightarrow \infty} g(x) = 0$. In particular, we have $g(x) > 0$ and $h''(x) > 0$ for all $x \geq 0$.*

Proof. Since $g(x) > 0$ and $g'(x) < 0$ for all $x > 0$, then $\lim_{x \longrightarrow \infty} g(x) = A$ for some constant $A \geq 0$. Now we only need to show that

$$A = 0.$$

If this were not true, we should have $A > 0$, which implies that $\lim_{x \longrightarrow \infty} \frac{1}{g(x)} = \frac{1}{A}$. Since $-\frac{h'''}{h''} = h + \beta + \frac{h'}{g}$, $x > 0$, and $\lim_{x \longrightarrow \infty} h(x) = \lim_{x \longrightarrow \infty} h'(x) = 0$, we have $\lim_{x \longrightarrow \infty} -\frac{h'''(x)}{h''(x)} = \beta$. Since $\lim_{x \longrightarrow \infty} h'(x) = \lim_{x \longrightarrow \infty} h''(x) = 0$, by L'Hospital's rule, we have

$$A = \lim_{x \longrightarrow \infty} g(x) = \lim_{x \longrightarrow \infty} -\frac{h''(x)}{h'(x)} = \lim_{x \longrightarrow \infty} -\frac{h'''(x)}{h''(x)} = \beta > 0.$$

Let's consider the function $c(x) := h(x) + \beta - 2g(x)$, $x > 0$, since $\lim_{x \longrightarrow \infty} h(x) = 0$ and $\lim_{x \longrightarrow \infty} g(x) = \beta$, we obtain $\lim_{x \longrightarrow \infty} c(x) = -\beta < 0$, which implies that there exists some $R_2 > 0$ such that $c(x) \leq -\frac{\beta}{2}$ for all $x \geq R_2$. The second equation (3.8) implies that $g'(x) = e^{-\int_{R_2}^x c(t)\, dt} g'(R_2)$ for all $x \geq R_2$. Since $g'(x) < 0$ for all $x \geq 0$, and $c(x) \leq -\frac{\beta}{2}$ for all $x \geq R_2$, then $g'(x) \leq g'(R_2) e^{\int_{R_2}^x \frac{\beta}{2}\, dt} = g'(R_2) e^{\frac{\beta}{2}(x - R_2)}$ for all $x \geq R_2$. Since $\beta > 0$ and $g'(R_2) < 0$, we have $\lim_{x \longrightarrow \infty} g'(x) = -\infty$, which contradicts with the limit $\lim_{x \longrightarrow \infty} g(x) = \beta$. $\square$

Corollary 3.1. *If $g'(x) < 0$ for all $x \geq 0$, there holds that $\lim_{x \longrightarrow \infty} g^{(k)}(x) = 0$ for all $k \in \mathbb{N}_0$.*

Proof. The statement follows directly from **Lemma 3.5**, **Proposition 3.2**, the equations in (3.8), and differentiating the equations in (3.8). $\square$

Lemma 3.6. *If $g'(x) < 0$ for all $x \geq 0$. For any $\gamma > \beta \geq 0$, then there exists some constant $C > 0$ such that*

$$h''(x) \geq Ce^{-\gamma x} \qquad \text{for all } x \geq 0.$$

In particular, since $\lim_{x\longrightarrow\infty} h'(x) = \lim_{x\longrightarrow\infty} h''(x) = 0$, *integrating the above inequality, we have*

$$-h'(x) \geq \frac{C}{\gamma}e^{-\gamma x} \quad \text{and} \quad h(x) \geq \frac{C}{\gamma^2}e^{-\gamma x} \qquad \text{for all } x \geq 0.$$

Proof. Rewriting the equation (3.1), we have

$$h_2' + (h+\beta)h_2 = (h')^2 \geq 0 \qquad \text{for all } x \geq 0.$$

For any $\gamma > \beta$ and fix it, consider the function $f(x) := e^{-\gamma x}$, $x \geq 0$, it's easy to see that

$$f' + (h+\beta)f = -\gamma e^{-\gamma x} + (h+\beta)e^{-\gamma x} = (h+\beta-\gamma)e^{-\gamma x} \qquad \text{for all } x \geq 0.$$

Since $\lim_{x\longrightarrow\infty} h(x) = 0$ and $\gamma > \beta$, there is some $R_2 > 0$ such that $h(x) + \beta - \gamma < 0$ for all $x \geq R_2$, which implies that

$$f' + (h+\beta)f < 0 \qquad \text{for all } x \geq R_2.$$

Consider the function $H(x) := h_2(x) - \frac{h_2(R_2)}{f(R_2)}f(x)$, $x \geq R_2$, since $h_2(R_2) = h''(R_2) > 0$ and $f(R_2) = e^{-\gamma R_2} > 0$, it's easy to see that

$$H(R_2) = 0, \qquad \text{and} \qquad H' + (h+\beta)H > 0 \quad \text{for all } x \geq R_2.$$

By **Lemma 1**, we have $H(x) \geq 0$ for all $x \geq R_2$, that is, $h''(x) \geq \frac{h_2(R_2)}{f(R_2)}f(x) = \frac{h_2(R_2)}{f(R_2)}e^{-\gamma x}$ for all $x \geq R_2$. Since $h_2(x) = h''(x) > 0$ for all $x \geq 0$, and $f(R_2) = e^{-\gamma R_2} > 0$, then there exists some constant $C > 0$ such that $h''(x) \geq Ce^{-\gamma x}$ for all $x \geq 0$. □

Lemma 3.7. *If* $g'(x) < 0$ *for all* $x \geq 0$. *If* $\beta > 0$, *then for any* $0 < \gamma < \beta$, *there exists some constant* $C > 0$ *such that*

$$-g'(x) \leq Ce^{-\gamma x} \qquad \text{for all } x \geq 0.$$

In particular, since $\lim_{x\longrightarrow\infty} g(x) = 0$, *integrating the above inequality, we have*

$$g(x) \leq \frac{C}{\gamma}e^{-\gamma x} \qquad \text{for all } x \geq 0.$$

Proof. Let $g_1(x) := -g'(x)$, $x \geq 0$, we have $g_1(x) > 0$ for all $x \geq 0$. The second equation in (3.8) says that

$$g_1' + (h+\beta-2g)g_1 = 0 \qquad \text{for all } x \geq 0.$$

For any $0 < \gamma < \beta$, consider the function $f(x) := e^{-\gamma x}$, $x \geq 0$, then

$$\begin{aligned} f' + (h + \beta - 2g) f &= -\gamma e^{-\gamma x} + (h + \beta - 2g) e^{-\gamma x} \\ &= (h + \beta - \gamma - 2g) e^{-\gamma x} \qquad \text{for all } x \geq 0. \end{aligned}$$

Since $\lim_{x \longrightarrow \infty} h(x) = \lim_{x \longrightarrow \infty} g(x) = 0$ and $0 < \gamma < \beta$, we have $\lim_{x \longrightarrow \infty} (h(x) + \beta - \gamma - 2g(x)) = \beta - \gamma > 0$. Hence there exists some $R_2 > 0$ such that $h(x) + \beta - \gamma - 2g(x) > 0$ for all $x \geq R_2$, which implies that

$$f' + (h + \beta - 2g) f > 0 \qquad \text{for all } x \geq R_2.$$

Consider the function $H(x) := \frac{g_1(R_2)}{f(R_2)} f(x) - g_1(x)$, $x \geq R_2$, since $g_1(R_2) = -g'(R_2) > 0$ and $f(R_2) = e^{-\gamma R_2} > 0$, it's easy to see that

$$H(R_2) = 0, \qquad \text{and} \qquad H' + (h + \beta - 2g) H > 0 \quad \text{for all } x \geq R_2.$$

By **Lemma 1**, we have $H(x) \geq 0$ for all $x \geq R_2$, that is, $\frac{g_1(R_2)}{f(R_2)} e^{-\gamma x} = \frac{g_1(R_2)}{f(R_2)} f(x) \geq g_1(x) = -g'(x)$ for all $x \geq R_2$. Since $g_1(x) = -g'(x) > 0$ for all $x \geq 0$, and $f(R_2) = e^{-\gamma R_2} > 0$, then there exists some constant $C > 0$ such that $-g'(x) \leq C e^{-\gamma x}$ for all $x \geq 0$. □

Lemma 3.8. *If $g'(x) < 0$ for all $x \geq 0$. If $\beta > 0$, then for any $0 < \gamma < \beta$, there exists some constant $C > 0$ such that*

$$h''(x) \leq C e^{-\gamma x} \qquad \text{for all } x \geq 0.$$

In particular, since $\lim_{x \longrightarrow \infty} h(x) = \lim_{x \longrightarrow \infty} h'(x) = 0$, integrating the above inequality, we have

$$-h'(x) \leq \frac{C}{\gamma} e^{-\gamma x} \text{ and } h(x) \leq \frac{C}{\gamma^2} e^{-\gamma x} \qquad \text{for all } x \geq 0.$$

Proof. For any $0 < \gamma < \beta$, by **Lemma 3.7**, then there exists some constant $C_1 > 0$ such that $-\frac{h''(x)}{h'(x)} = g(x) \leq C_1 e^{-\gamma x}$ for all $x \geq 0$. Since $h'(x) < 0$ for all $x \geq 0$, we have $h''(x) \leq -C_1 e^{-\gamma x} h'(x)$ for all $x \geq 0$. Since $h''(x) > 0$ for all $x \geq 0$, we obtain $-h'(x) \leq -h'(0)$ for all $x \geq 0$, which implies that $h''(x) \leq -C_1 e^{-\gamma x} h'(0)$ for all $x \geq 0$. Let $C := -C_1 h'(0)$, since $h'(x) < 0$ for all $x \geq 0$, we have $C > 0$ and $h''(x) \leq C e^{-\gamma x}$ for all $x \geq 0$. □

Lemma 3.9. *If $g'(x) < 0$ for all $x \geq 0$, we must have $\beta = 0$.*

Proof. If the statement is not true, by **Proposition 3.3**, we should have $\beta > 0$, which implies that $0 < \frac{3\beta}{4} < \beta < \frac{4\beta}{3}$. By **Lemma 3.6**, **Lemma 3.7** and **Lemma 3.8**, then there are some constants $C_1 > 0$, $C_2 > 0$, $C_3 > 0$ such that

$$h''(x) \geq C_1 e^{-\frac{4\beta}{3} x}, \quad 0 < g(x) \leq C_2 e^{-\frac{3\beta}{4} x}, \quad \text{and} \quad 0 < -h'(x) \leq C_3 e^{-\frac{3\beta}{4}} \qquad \text{for all } x \geq 0.$$

Since $h''(x) = -g(x)h'(x)$ for all $x \geq 0$, we get $C_1 e^{-\frac{4\beta}{3}x} \leq C_2 C_3 e^{-\frac{3\beta}{2}x}$ for all $x \geq 0$, that is, $e^{\frac{\beta}{6}x} \leq \frac{C_2 C_3}{C_1}$ for all $x \geq 0$, which implies that $\beta \leq 0$, contradiction. □

Corollary 3.2. *There holds that $\beta = 0$ if and only if $g'(x) < 0$ for all $x \geq 0$.*

Proof. The statement follows directly from the alternative in (3.10), **Proposition 3.4**, and **Lemma 3.9**. □

Corollary 3.3. *Assume that $\beta > 0$, let h be a non-constant bounded solution of the equation (3.1) with $h'(0) \leq 0$ and $\lim_{x \longrightarrow \infty} h(x) = 0$, then there is some constant $a > 0$ such that $h(x) = ae^{-\beta x}$, $x \geq 0$.*

Proof. The statement follows directly from **Corollary 3.2**, the alternative in (3.10) and **Proposition 3.4**. □

Proposition 3.5. *If $\beta = 0$, then we have $h'' + hh' < 0$ and $h' + \frac{1}{2}h^2 > 0$ for all $x \geq 0$, and $h(x) \geq \frac{1}{\frac{x}{2} + \frac{1}{h(0)}}$ for all $x \geq 0$. In particular, since $g(x) = -\frac{h''(x)}{h'(x)}$ for all $x \geq 0$, we have $0 < g(x) < h(x)$ for all $x \geq 0$.*

Proof. Since $\beta = 0$, the equation (3.1) becomes $h''' + hh'' = (h')^2$, $x \geq 0$. Hence we have $(h'' + hh')' = h''' + hh'' + (h')^2 = 2(h')^2 > 0$, $x \geq 0$. Since $\lim_{x \longrightarrow \infty} h(x) = \lim_{x \longrightarrow \infty} h'(x) = \lim_{x \longrightarrow \infty} h''(x) = 0$, we obtain $\left(h' + \frac{1}{2}h^2\right)' = h'' + hh' < 0$ for all $x \geq 0$. Since $\lim_{x \longrightarrow \infty} h(x) = \lim_{x \longrightarrow \infty} h'(x) = 0$, we have $h' + \frac{1}{2}h^2 > 0$ for all $x \geq 0$. Since $h(x) > 0$ for all $x \geq 0$, we have $\frac{h'}{h^2} > -\frac{1}{2}$ for all $x \geq 0$, integrating the inequality gives us that $-\frac{1}{h(x)} + \frac{1}{h(0)} \geq -\frac{x}{2}$ for all $x \geq 0$. Since $h(x) > 0$ for all $x \geq 0$, we obtain $h(x) \geq \frac{1}{\frac{x}{2} + \frac{1}{h(0)}}$ for all $x \geq 0$. □

Proposition 3.6. *If $\beta = 0$, there holds that $h''' = (h')^2 - hh'' < 0$ for all $x \geq 0$. In particular, we have $\left(-\frac{h'}{h}\right)' = \frac{(h')^2 - hh''}{h^2} = \frac{h'''}{h^2} < 0$ for all $x \geq 0$.*

Proof. Since $\beta = 0$, by **Corollary 3.2**, we have $g'(x) < 0$ for all $x \geq 0$, and the equation (3.1) becomes $h''' + hh'' = (h')^2$, $x \geq 0$. By the definition of g and the first equation in (3.8) with $\beta = 0$, together with the facts $h'(x) < 0$ and $g'(x) < 0$ for all $x \geq 0$, we have

$$h''' = (h')^2 - hh'' = (h')^2 + ghh' = h'(h' + gh) = h'(g^2 - g') < 0 \qquad \text{for all } x \geq 0. \quad \square$$

Theorem 3.1. *Let h be a non-constant bounded solution of the equation $h''' + hh'' = (h')^2$, $x \geq 0$, with $h'(0) \leq 0$.*

(a) *Then $h'(x) < 0$ for all $x \geq 0$. In particular, the limit $\beta := \lim_{x \longrightarrow \infty} h(x)$ exists.*

(b) *Then $\beta \geq 0$. Moreover if $\beta > 0$, then there exists some constant $a > 0$ such that $h(x) = ae^{-\beta x} + \beta$, $x \geq 0$; if $\beta = 0$, then $(-1)^k h^{(k)}(x) > 0$, $x \geq 0$, for all $k = 1, 2, 3$.*

Proof. (a) The statement follows directly from **Lemma 3.1**, **Proposition 3.4** and (3.7). (b) Let $H(x) := h(x) - \beta$, $x \geq 0$, then $H''' + (H+\beta)H'' = (H')^2$ for all $x \geq 0$, $H'(0) = h'(0) \leq 0$, and $\lim_{x \longrightarrow \infty} H(x) = 0$. By **Proposition 3.3**, we have $\beta \geq 0$. Moreover, if $\beta > 0$, by **Corollary 3.3**, then there exists some constant $a > 0$ such that $H(x) = ae^{-\beta x}$, $x \geq 0$, which implies that $h(x) = ae^{-\beta x} + \beta$, $x \geq 0$. If $\beta = 0$, by **Corollary 3.2**, **Lemma 3.5**, (3.7), and **Proposition 3.6**, we have $(-1)^k H^{(k)}(x) > 0$, $x \geq 0$, for all $k = 1, 2, 3$, which implies that $(-1)^k h^{(k)}(x) > 0$, $x \geq 0$, for all $k = 1, 2, 3$. □

Remark 3.1. Consider the space $X := \{u \in C^3([0,\infty)) \mid u(0) = 0 \text{ and } \|(1+x)\cdot u(x)\|_{L^\infty([0,\infty))} < \infty\}$, for any $u \in X$, let's plug $h(x) := \frac{6}{x+a} + u(x)$ into the equation $h''' + hh'' = (h')^2$, we have

$$-\frac{36}{(x+a)^4} + u''' + \left(\frac{6}{x+a} + u\right)\left(\frac{12}{(x+a)^3} + u''\right) = \left(-\frac{6}{(x+a)^2} + u'\right)^2, \qquad \text{for all } x \geq 0.$$

Simplifying the above equation gives us that

$$Lu := u''' + \left(\frac{6}{x+a} + u\right)u'' + \left(\frac{12}{(x+a)^2} - u'\right)u' + \frac{12}{(x+a)^3}u = 0, \qquad \text{for all } x \geq 0.$$

Let $F(t) := L(tu)$, $t \in \mathbb{R}$, we have

$$F(t) = tu''' + t\left(\frac{6}{x+a} + tu\right)u'' + t\left(\frac{12}{(x+a)^2} - tu'\right)u' + \frac{12t}{(x+a)^3}u, \qquad \text{for all } u \in X.$$

Hence we have

$$L'(0)[u] = u''' + \frac{6}{x+a}u'' + \frac{12}{(x+a)^2}u' + \frac{12}{(x+a)^3}u, \qquad \text{for all } u \in X.$$

Claim I: $\ker(L'(0)) = \{0\}$.

For any $u \in \ker(L'(0))$, we have $u''' + \frac{6}{x+a}u'' + \frac{12}{(x+a)^2}u' + \frac{12}{(x+a)^3}u = 0$, $x \geq 0$, that is, we have

$$(x+a)^3u''' + 6(x+a)^2u'' + 12(x+a)u' + 12u = 0, \qquad \text{for all } x \geq 0. \tag{3.11}$$

Hence the equation (3.11) is a Cauchy-Euler equation, its characteristic equation is $r(r-1)(r-2) + 6r(r-1) + 12r + 12 = 0$, that is, $(r+2)(r^2+r+6) = 0$, which implies that there are some constants $A, B, C \in \mathbb{R}$ such that

$$u(x) = \frac{A}{(x+a)^2} + \frac{B}{\sqrt{x+a}}\cos\left(\frac{\sqrt{23}}{2}\ln(x+a)\right) + \frac{C}{\sqrt{x+a}}\sin\left(\frac{\sqrt{23}}{2}\ln(x+a)\right), \qquad \text{for all } x \geq 0.$$

Since $u \in X$, we know that u has linear decay at infinity, we must have $B = C = 0$, that is, $u(x) = \frac{A}{(x+a)^2}$. Since $u \in X$, we have $u(0) = 0$, hence we obtain $A = 0$, that is, $u(x) \equiv 0$, $x \geq 0$. Therefore we can conclude that $\ker(L'(0)) = \{0\}$.

Hence **Claim I** tells us that the solution $y_a(x) = \frac{6}{x+a}$ is a non-degenerate solution to the equation $h''' + hh'' = (h')^2$, $x \geq 0$. By using the Implicit Function Theorem, we know that near the solution $y_a(x) = \frac{6}{x+a}$ there are no other solutions in the space X to the equation $h''' + hh'' = (h')^2$, $x \geq 0$.

Conjecture 3.1. *Let h be a non-constant bounded solution of the equation $h''' + hh'' = (h')^2$, $x \geq 0$, such that $h'(0) \leq 0$ and $\lim_{x \longrightarrow \infty} h(x) = 0$, then there is some constant $a > 0$ such that $h(x) = \frac{6}{x+a}$ for all $x \geq 0$.*

3.2. Bounded solutions on the right half line when $h'(0) > 0$

In this subsection let's assume that $h(x)$ is a bounded solution of equation (3.1) with the condition $h'(0) > 0$. In this case, eventually we are going to show that after possibly adding a constant to $h(x)$ such that the limit of $h(x)$ at infinity is 0, we have $\beta > 0$ and $h(x) = ae^{-\beta x}$ for some constant $a < 0$.

Proposition 3.7. *There holds that $h'(x) > 0$ for all $x \geq 0$. In particular, since h is a bounded function on $[0, \infty)$, we know that the limit $h(\infty) := \lim_{x \longrightarrow \infty} h(x)$ exists.*

Proof. If the assertion is not true, since $h'(0) > 0$, we know that there exists some $b > 0$ such that $h'(x) > 0 = h'(b)$ for all $0 \leq x < b$. Since h is a non-constant bounded solution of equation (3.1), by the theorem of uniqueness for solutions to the ODE, we know that h can not be a constant function in any open subinterval of $[0, \infty)$. Let $\phi(x) := h(x + b)$ for all $x \geq 0$, it's easy to see that ϕ is a bounded solution of the equation (3.1), and ϕ can not be a constant function in any open subinterval of $[0, \infty)$. Since $\phi'(0) = h'(b) = 0$, by the analysis in **Subsection 3.1**, we get $\phi'(x) < 0$ for all $x \geq 0$, contradiction. □

By considering $h(x) - h(\infty)$, W.L.O.G., in the following of this subsection let's assume that

$$h(\infty) = \lim_{x \longrightarrow \infty} h(x) = 0. \tag{3.12}$$

Proposition 3.8. *There holds that $h''(x) < 0$ for all $x \geq 0$.*

Proof. If the statement is not true, that is, there exists some $a \geq 0$ such that $h_2(a) = h''(a) \geq 0$. Since $h_2' + (h + \beta)h_2 = (h')^2 \geq 0$, $x \geq 0$, by **Lemma 1**, we have $h''(x) \geq 0$ for all $x \geq a$. Hence $h'(x) \geq h'(a)$ for all $x \geq a$, which implies that $h(x) \geq h'(a)(x - a) + h(a)$ for all $x \geq a$. Since $h'(a) > 0$, we have $\lim_{x \longrightarrow \infty} h(x) = \infty$, contradiction. □

Since $\lim_{x \longrightarrow \infty} h(x) = 0$ and $h''(x) < 0$ for all $x \geq 0$, by the mean value theorem, we have

$$\lim_{x \longrightarrow \infty} h'(x) = 0. \tag{3.13}$$

Proposition 3.9. *There holds that* $\lim_{x\longrightarrow\infty} h''(x) = 0$.

Proof. By the same argument as **Proposition 3.1** with the help of the limits (3.12) and (3.13). □

Lemma 3.10. *There holds that* $h_2(x) = \frac{h_2(0)}{h_1(0)} h_1(x)$ *for all* $x \geq 0$.

Proof. Let's define the function $f(x) := h_2(x) - \frac{h_2(0)}{h_1(0)} h_1(x)$ for all $x \geq 0$, by the equations (2) and (3), we know that

$$f'' + (h+\beta) f' - h' f = 0 \qquad \text{for all } x \geq 0.$$

Claim I: There holds that $f(x) \equiv 0$ for all $x \geq 0$. In particular, we have $h_2(x) = \frac{h_2(0)}{h_1(0)} h_1(x)$ for all $x \geq 0$.

If the statement is not true, that is, $f(x) \not\equiv 0$ for all $x \geq 0$, since $f(0) = 0$, then there exists some $a > 0$ such that $f(a) \neq 0$. By considering $\pm f(x)$, W.L.O.G., let's assume that $f(a) > 0$. Since $\lim_{x\longrightarrow\infty} f(x) = 0$, then there exists some $b > 0$ such that b is a global interior maximum point of $f(x)$, which implies that

$$f(b) > 0, \qquad f'(b) = 0, \qquad \text{and} \qquad f''(b) \leq 0.$$

Hence we have $0 = f''(b) + (h(b)+\beta) f'(b) - h'(b) f(b) = f''(b) - h'(b) f(b) \leq -h'(b) f(b)$, which contradicts with $h'(b) > 0$ and $f(b) > 0$. □

Proposition 3.10. *Let* h *be a non-constant bounded solution of equation* (3.1) *with the conditions* $h'(0) < 0$ *and* $\lim_{x\longrightarrow\infty} h(x) = 0$, *then we must have* $\beta > 0$ *and there is some constant* $a < 0$ *such that* $h(x) = ae^{-\beta x}$, $x \geq 0$.

Proof. By **Lemma 3.10**, **Proposition 3.7** and **Proposition 3.8**, then there exists some constant $\gamma > 0$ such that $h''(x) + \gamma h'(x) = 0$ for all $x \geq 0$. Hence there are some constants $a, b \in \mathbb{R}$ such that $h(x) = ae^{-\gamma x} + b$ for all $x \geq 0$. Since $\lim_{x\longrightarrow\infty} h(x) = 0$, $h'(x) > 0$ for all $x \geq 0$, and $\gamma > 0$, we have

$$b = 0, \qquad h(0) = a < 0, \qquad \text{and} \qquad h(x) = ae^{-\gamma x} \quad \text{for all } x \geq 0.$$

Let's plug $h(x) = ae^{-\gamma x}$ into the equation (1), we obtain $-a\gamma^3 e^{-\gamma x} + (ae^{-\gamma x} + \beta) \cdot a\gamma^2 e^{-\gamma x} = (-a\gamma e^{-\gamma x})^2$ for all $x \geq 0$, that is, $a\gamma^2 e^{-\gamma x}(-\gamma + \beta) = 0$ for all $x \geq 0$. Since $a < 0$ and $\gamma > 0$, we obtain $\beta = \gamma > 0$ and $h(x) = ae^{-\beta x}$ for all $x \geq 0$. □

Theorem 3.2. *Let* h *be a non-constant bounded solution of the equation* $h''' + hh'' = (h')^2$, $x \geq 0$, *with the condition* $h'(0) > 0$, *then there are some constants* $a < 0$ *and* $\beta > 0$ *such that* $h(x) = ae^{-\beta x} + \beta$, $x \geq 0$.

Proof. By **Proposition 3.7**, we have $h'(x) > 0$ for all $x \geq 0$, which implies that the limit $\beta := \lim_{x\longrightarrow-\infty} h(x)$ exists. Let $H(x) := h(x) - \beta$, $x \geq 0$, then $H''' + (H+\beta)H'' = (H')^2$ for all $x \geq 0$,

$H'(0) = h'(0) > 0$, and $\lim_{x\longrightarrow\infty} H(x) = 0$. By **Proposition 3.10**, we have $\beta > 0$, and there exists some constant $a < 0$ such that $H(x) = ae^{-\beta x}$, $x \geq 0$, which implies that $h(x) = ae^{-\beta x} + \beta$, $x \geq 0$. □

Corollary 3.4. *Let h be a non-constant bounded solution of the equation $h''' + (h+\beta)h'' = (h')^2$, $x \geq 0$, with the condition $h'(0) > 0$, then there are some constants $a < 0$ and $\gamma > 0$ such that $h(x) = ae^{-\gamma x} + \gamma - \beta$, $x \geq 0$.*

Proof. Let $H(x) := h(x) + \beta$, $x \geq 0$, then $H''' + HH'' = (H')^2$ for all $x \geq 0$, and $H'(0) = h'(0) > 0$. By **Theorem 3.2**, then there are some constants $a < 0$ and $\gamma > 0$ such that $H(x) = ae^{-\gamma x} + \gamma$, $x \geq 0$, which implies that $h(x) = ae^{-\gamma x} + \gamma - \beta$, $x \geq 0$. □

3.3. Bounded solutions on the whole line

To study the bounded solution of the equation (3.1), notice that the function $H(x) := -(h(-x) + \beta)$ satisfies the equation $H''' + HH'' = (H')^2$ provided $h(x)$ is a solution of the equation (3.1), so when applying **Theorem 3.1** and **Theorem 3.2** to the function $H(x)$, we can obtain the properties of the bounded solution of the equation (3.1) on the left half line $(-\infty, 0]$.

Theorem 3.3. *Let h be a non-constant bounded solution of the equation $h''' + hh'' = (h')^2$, $x \leq 0$.*

(a) *If $h'(0) \leq 0$, then $h'(x) < 0$ for all $x \leq 0$. In particular, the limit $\beta := \lim_{x\longrightarrow-\infty} h(x)$ exists, and $\beta \leq 0$. If further assume that $\beta < 0$, then there exists constant $a < 0$ such that $h(x) = ae^{-\beta x} + \beta$, $x \leq 0$; if further assume that $\beta = 0$, then $h^{(k)}(x) < 0$, $x \leq 0$, for all $k = 0, 1, 2, 3$.*
(b) *If $h'(0) > 0$, then there are some constants $a > 0$ and $\beta < 0$ such that $h(x) = ae^{-\beta x} + \beta$, $x \leq 0$.*

For any bounded solution $h(x)$ of the equation (1) on the whole line, applying **Theorem 3.1**, **Theorem 3.2**, **Theorem 3.3**, together with the theorem of uniqueness for solutions to the ODE, to the function $h(x) + \beta$, we can obtain the following Liouville type result.

Theorem 3.4. *The only bounded solutions of the equation (1) on the whole line are constant functions.*

Proof. If the statement is not true, by considering the function $h(x) + \beta$, let us assume that $h(x)$ is a non-constant bounded solution of the equation $h''' + hh'' = (h')^2$, $x \in \mathbb{R}$.

If $h'(0) \leq 0$, by **Theorem 3.1** and **Theorem 3.3**, we have $h'(x) < 0$ for all $x \in \mathbb{R}$, and there are some constants $A, B \in \mathbb{R}$ such that $\lim_{x\longrightarrow\infty} h(x) = A \geq 0$, and $\lim_{x\longrightarrow-\infty} h(x) = B \leq 0$, which contradicts with the fact that $h'(x) < 0$ for all $x \in \mathbb{R}$.

If $h'(0) > 0$, by **Theorem 3.2** and **Theorem 3.3**, then there are some constants $a_1 < 0$, $a_2 > 0$, $\beta_1 > 0$ and $\beta_2 < 0$ such that $h(x) = a_1 e^{-\beta_1 x} + \beta_1$ for all $x \geq 0$, and $h(x) = a_2 e^{-\beta_2 x} + \beta_2$ for all $x \leq 0$, which implies that $h''(0) = a_1 \cdot \beta_1^2 < 0$ and $h''(0) = a_2 \cdot \beta_2^2 > 0$, contradiction. □

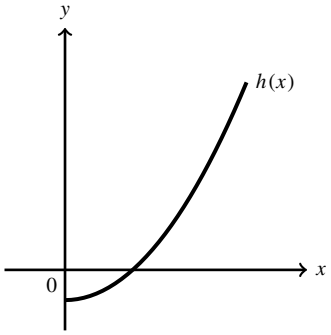


Fig. 3. $h'(0) \geq 0$ and $h''(0) \geq 0$.

4. Unbounded solutions

In this section we are interesting in the unbounded solutions of the equation (1), by shifting the function via $h(x) \longmapsto h(x) + \beta$, we may assume without loss of generality that $\beta = 0$. By considering $-h(-x)$, we only need to focus the equation (1) on the right half line $[0, \infty)$. Thus, we now consider the following nonlinear ordinary differential equation:

$$h''' + hh'' = (h')^2 \qquad \text{for all } x \geq 0. \tag{4.1}$$

4.1. Unbounded solutions when $h'(0) \geq 0$ and $h''(0) \geq 0$

In this subsection, we assume $h(x)$ is a non-constant solution to equation (4.1) satisfying $h'(0) \geq 0$ and $h''(0) \geq 0$. We will show that under these conditions, $h(x)$ is strictly increasing, concave up, and grows exponentially as $x \longrightarrow \infty$.

Proposition 4.1. *There holds that $h''(x) > 0$ for all $x > 0$. In particular, since $h'(0) \geq 0$, we have $h'(x) > 0$ for all $x > 0$, and $\lim_{x \longrightarrow \infty} h(x) = \infty$ (Fig. 3).*

Proof. The statement follows directly from **Proposition 2.1**. □

Lemma 4.1. *There exists some $a_1 \geq 0$ such that $h'''(a_1) > 0$.*

Proof. If the statement is not true, we have

$$h'''(x) \leq 0 \qquad \text{for all } x \geq 0. \tag{4.2}$$

Claim I: $h'''(x) < 0$ for all $x \geq 0$.

If not, that is, there exists some $a \geq 0$ such that $h'''(a) = 0$. Since $h'(x) > 0$ and $h''(x) > 0$ for all $x > 0$, we obtain $h_3' + hh_3 = h'h'' \geq 0$, $x \geq 0$. By **Lemma 1**, we have $h'''(x) \geq 0$ for all $x \geq a$. Combine this with (4.2), we obtain $h'''(x) \equiv 0$ for all $x \geq a$, that is, $h(x)$ is quadratic polynomial for $x \geq a$. Since only quadratic polynomial solutions of the equation (4.1) are constant functions, this contradicts with the fact that $h'(x) > 0$ for all $x > 0$.

Since $h'(x) > 0$ and $h''(x) > 0$ for all $x > 0$, by **Claim I**, we know that there exists some constants $A \geq 0$ and $B \geq 0$ such that

$$\lim_{x\longrightarrow\infty} h''(x) = A, \qquad \text{and} \qquad \lim_{x\longrightarrow\infty} \frac{1}{h'(x)} = B.$$

If $B > 0$ we must have $\lim_{x\longrightarrow\infty} h'(x) = \frac{1}{B}$. By the mean value theorem, since $\lim_{x\longrightarrow\infty} h''(x) = A$, we must have $A = 0$. Hence we must have $AB = 0$, which implies that

$$\lim_{x\longrightarrow\infty} \frac{h''(x)}{h'(x)} = 0. \tag{4.3}$$

Since $h'(x) > 0$ and $h''(x) > 0$ for all $x > 0$, we can define the function $g(x) := -\frac{h''(x)}{h'(x)}$, $x > 0$, so we obtain

$$g(x) < 0 \quad \text{for all } x > 0, \qquad \text{and} \qquad \lim_{x\longrightarrow\infty} g(x) = 0.$$

It's easy to see that

$$g' + hg + h' - g^2 = 0, \quad \text{and} \quad g'' + (h - 2g)g' = 0 \qquad \text{for all } x > 0. \tag{4.4}$$

Since the derivative $g'(x)$ solves the linear equation $y' + (h - 2g)y = 0$, $x > 0$, by the theorem of uniqueness for solutions to the ODE, we have

$$\text{either} \qquad g'(x) \equiv 0 \text{ for all } x > 0, \qquad \text{or} \qquad g'(x) > 0 \text{ for all } x > 0, \qquad \text{or}$$
$$g'(x) < 0 \text{ for all } x > 0.$$

Since $g(x) < 0$ for all $x > 0$ and $\lim_{x\longrightarrow\infty} g(x) = 0$, we must have $g'(x) > 0$ for all $x > 0$. Since $g'' = -(h - 2g)g'$, $x > 0$, since $\lim_{x\longrightarrow\infty} h(x) = \infty$, and $g(x) < 0$ and $g'(x) > 0$ for all $x > 0$, there exists some large $R_2 \gg 1$ such that

$$h(x) > 0 \quad \text{and} \quad g''(x) < 0 \qquad \text{for all } x \geq R_2.$$

Since $\lim_{x\longrightarrow\infty} g(x) = 0$, by the mean value theorem, we have $\lim_{x\longrightarrow\infty} g'(x) = 0$. Since $\lim_{x\longrightarrow\infty} g(x) = 0$, by the first equation in (4.4), we have

$$\lim_{x\longrightarrow\infty} h \cdot \left(g + \frac{h'}{h}\right) = \lim_{x\longrightarrow\infty} (hg + h') = \lim_{x\longrightarrow\infty} (-g' + g^2) = 0.$$

Since $\lim_{x\longrightarrow\infty} h(x) = \infty$, we must have $\lim_{x\longrightarrow\infty} \left(g + \frac{h'}{h}\right) = 0$. Since $\lim_{x\longrightarrow\infty} g(x) = 0$, we have $\lim_{x\longrightarrow\infty} \frac{h'}{h} = 0$. Since $h(x) > 0$ for all $x \geq R_2$, we can define the function $u(x) := \frac{h'(x)}{h(x)}$ for all $x \geq R_2$, since $h'(x) > 0$ for all $x > 0$, we obtain

$$u(x) > 0 \quad \text{for all } x \geq R_2, \qquad \text{and} \qquad \lim_{x\longrightarrow\infty} u(x) = 0. \tag{4.5}$$

On the other hand, by the equation (4.1) and **Claim I**, we have $u'(x) = \frac{h''\cdot h-(h')^2}{h^2} = -\frac{h'''}{h^2} > 0$, $x \geq R_2$, which contradicts with (4.5). □

Proposition 4.2.

(a) *If $h'''(0) \geq 0$, that is, $(h'(0))^2 \geq h(0)h''(0)$, then there holds that $h'''(x) > 0$ for all $x > 0$.*
(b) *If $h'''(0) < 0$, that is, $(h'(0))^2 < h(0)h''(0)$, then there exists some $a > 0$ such that $h'''(x) < 0$ for all $0 \leq x < a$ and $h'''(x) > 0$ for all $x > a$.*

Proof. Since $h'(x) > 0$ and $h''(x) > 0$ for all $x > 0$, we have

$$h_3' + hh_3 = h'h'' > 0 \qquad \text{for all } x > 0.$$

(a) If $h'''(0) \geq 0$, by **Lemma 1**, we have $h'''(x) > 0$ for all $x > 0$.
(b) If $h'''(0) < 0$, by **Lemma 4.1**, then there is some $0 < a < a_1$ such that $h'''(0) < 0 = h'''(a)$ for all $0 \leq x < a$. By **Lemma 1** again, we obtain $h'''(x) > 0$ for all $x > a$. □

Corollary 4.1. *There holds that* $\lim_{x\longrightarrow\infty} h'(x) = \infty$.

Proof. The statement follows directly from the facts $h'(x) > 0$ and $h''(x) > 0$ for all $x > 0$, and $h'''(x) > 0$ for $x \gg 1$. □

Proposition 4.3. *There exists some positive constant $A > 0$ such that $\lim_{x\longrightarrow\infty} \frac{h''(x)}{h'(x)} = A$. In particular, since $\lim_{x\longrightarrow\infty} h(x) = \lim_{x\longrightarrow\infty} h'(x) = \infty$, by using L'Hospital's rule, we obtain $\lim_{x\longrightarrow\infty} \frac{h'(x)}{h(x)} = \lim_{x\longrightarrow\infty} \frac{\ln h(x)}{x} = A$. As a consequence, we know that $h(x)$ approaches to infinity in an exponential rate as x goes to infinity.*

Proof. By **Proposition 4.1** and **Proposition 4.2**, up to a right translation, W.L.O.G., we may assume that

$$h^{(k)}(x) > 0 \quad \text{for all } x \geq 0, \qquad \text{for all } k = 0, 1, 2, 3.$$

Let's consider the function $g(x) := -\frac{h''(x)}{h'(x)}$ for all $x \geq 0$, we have

$$g(x) < 0, \qquad g' + hg + h' - g^2 = 0 \quad \text{and} \quad g'' + (h - 2g)g' = 0 \qquad \text{for all } x \geq 0. \tag{4.6}$$

Since the derivative $g'(x)$ solves the linear equation $y' + (h-2g)y = 0$, $x \geq 0$, by the theorem of uniqueness for solutions to the ODE, we have

$$\begin{aligned} &\text{either} \qquad g'(x) \equiv 0 \text{ for all } x \geq 0, \qquad \text{or} \qquad g'(x) > 0 \text{ for all } x \geq 0, \\ &\text{or} \qquad g'(x) < 0 \text{ for all } x \geq 0. \end{aligned} \tag{4.7}$$

Claim I: If $g'(x) \le 0$ for all $x \ge 0$, then there exists some positive constant $A > 0$ such that $\lim\limits_{x\longrightarrow\infty} \dfrac{h''(x)}{h'(x)} = A$.

Since $g(x) < 0$ and $g'(x) \le 0$ for all $x \ge 0$, then $\lim\limits_{x\longrightarrow\infty} \dfrac{h''(x)}{h'(x)} = \lim\limits_{x\longrightarrow\infty} -g(x) = A$ for some $A \in (0, \infty]$. Now we only need to show that

$$A < \infty.$$

If this is not true, that is, $A = \lim\limits_{x\longrightarrow\infty} -g(x) = \lim\limits_{x\longrightarrow\infty} \dfrac{h''(x)}{h'(x)} = \infty$. Since $g'(x) \le 0$ for all $x \ge 0$, by the alternative in (4.7), we have $g'(x) < 0$ for all $x \ge 0$. Since $g'' = -(h - 2g)g'$, and $h(x) > 0$, $g(x) < 0$ and $g'(x) < 0$ for all $x \ge 0$, we have $g''(x) \ge 0$ for all $x \ge 0$. Since $g'(x) < 0$ for all $x \ge 0$, hence there exists some constant $B \le 0$ such that $\lim\limits_{x\longrightarrow\infty} g'(x) = B$. By the mean value theorem, there exists an increasing sequence $\{x_n\}_{n=1}^{\infty} \subseteq (0, \infty)$ such that

$$\lim_{n\longrightarrow\infty} x_n = \infty, \qquad \text{and} \qquad \lim_{n\longrightarrow\infty} g''(x_n) = 0. \tag{4.8}$$

Since $g'' = -(h - 2g)g'$, $x \ge 0$, $\lim\limits_{x\longrightarrow\infty} h(x) = \infty$, $\lim\limits_{x\longrightarrow\infty} g(x) = -\infty$ and $\lim\limits_{x\longrightarrow\infty} g'(x) = B \le 0$, by (4.8), we must have $\lim\limits_{x\longrightarrow\infty} g'(x) = B = 0$. Since $g'' = -(h - 2g)g'$ and $g'(x) < 0$ for all $x \ge 0$, $\lim\limits_{x\longrightarrow\infty} h(x) = \infty$, and $\lim\limits_{x\longrightarrow\infty} g(x) = -\infty$, we have

$$\lim_{x\longrightarrow\infty} \frac{g''(x)}{g'(x)} = \lim_{x\longrightarrow\infty} -(h(x) - 2g(x)) = -\infty.$$

Since $\lim\limits_{x\longrightarrow\infty} g'(x) = 0$ and $g'(x) < 0$ for all $x \ge 0$, by L'Hospital's rule, we obtain $\lim\limits_{x\longrightarrow\infty} \dfrac{\ln(-g'(x))}{x} = \lim\limits_{x\longrightarrow\infty} \dfrac{g''(x)}{g'(x)} = -\infty$. Hence there exists some $R_2 > 0$ such that $\frac{\ln(-g'(x))}{x} \le -1$ for all $x \ge R_2$, that is, $-g'(x) \le e^{-x}$ for all $x \ge R_2$. Integrating this inequality yields that $g(R_2) - g(x) \le e^{-R_2} - e^{-x} \le e^{-R_2}$ for all $x \ge R_2$. So we obtain $-g(x) \le e^{-R_2} - g(R_2)$ for all $x \ge R_2$, which implies that $-g(x)$ is bounded from above on $[R_2, \infty)$. This contradicts with the fact that $\lim\limits_{x\longrightarrow\infty} -g(x) = \infty$. Hence we must have $0 < A < \infty$.

Claim II: If $g'(x) > 0$ for all $x \ge 0$, then there exists some positive constant $A > 0$ such that $\lim\limits_{x\longrightarrow\infty} \dfrac{h''(x)}{h'(x)} = A$.

Since $g(x) < 0$ and $g'(x) > 0$ for all $x \ge 0$, it's easy to know that there exist some constant $A \ge 0$ such that $\lim\limits_{x\longrightarrow\infty} \dfrac{h''(x)}{h'(x)} = \lim\limits_{x\longrightarrow\infty} -g(x) = A$. So it suffices to show that

$$A > 0.$$

If this is not true, that is, $A = \lim\limits_{x\longrightarrow\infty} g(x) = 0$. Since $g'' = -(h - 2g)g'$, and $h(x) > 0$, $g(x) < 0$ and $g'(x) > 0$ for all $x \ge 0$, we have $g''(x) < 0$ for all $x \ge 0$. Since $\lim\limits_{x\longrightarrow\infty} g(x) = 0$, by the mean

value theorem, we obtain $\lim_{x\longrightarrow\infty} g'(x) = 0$. Since $g(x) < 0$ for all $x \geq 0$ and $\lim_{x\longrightarrow\infty} g(x) = 0$, by L'Hospital's Rule and $g'' = -(h - 2g)g'$, $x \geq 0$, we have

$$\lim_{x\longrightarrow\infty} \frac{\ln(-g(x))}{x} = \lim_{x\longrightarrow\infty} \frac{g'(x)}{g(x)} = \lim_{x\longrightarrow\infty} \frac{g''(x)}{g'(x)} = \lim_{x\longrightarrow\infty} -(h(x) - 2g(x)) = -\infty.$$

So there exists some large $R_2 > 0$ such that $\frac{\ln(-g(x))}{x} \leq -1$ for all $x \geq R_2$, which implies that $(\ln h'(x))' = \frac{h''(x)}{h'(x)} = -g(x) \leq e^{-x}$ for all $x \geq R_2$. Integrating this inequality gives us that $\ln \frac{h'(x)}{h'(R_2)} \leq e^{-R_2} - e^{-x} \leq 1$ for all $x \geq R_2$, which implies that $h'(x) \leq h'(R_1)e$ for all $x \geq R_2$. This contradicts with the fact that $\lim_{x\longrightarrow\infty} h'(x) = \infty$.

In summary, we can conclude that there exists some positive constant $A > 0$ such that $\lim_{x\longrightarrow\infty} \frac{h''(x)}{h'(x)} = A$. □

Remark 4.1. If $g'(x) \equiv 0$ for all $x \geq 0$ in the proof of **Proposition 4.3**, then there exists some constant $\alpha > 0$ such that $h''(x) - \alpha h'(x) = 0$ for all $x \geq 0$. Hence there are some $a, b \in \mathbb{R}$ such that

$$h(x) = ae^{\alpha x} + b \qquad \text{for all } x \geq 0.$$

Since $\lim_{x\longrightarrow\infty} h(x) = \infty$, we must have $a > 0$ and $\alpha > 0$. Plugging $h(x) = ae^{\alpha x} + b$ into the equation (4.1), we have $a\alpha^3 e^{\alpha x} + (ae^{\alpha x} + b)a\alpha^2 e^{\alpha x} = (a\alpha e^{\alpha x})^2$ for all $x \geq 0$, that is, we have $e^{\alpha x}(a\alpha^3 + ab\alpha^2) = 0$ for all $x \geq 0$. Since $a > 0$ and $\alpha > 0$, we obtain $b = -\alpha$. Therefore we have

$$h(x) = ae^{\alpha x} - \alpha \qquad \text{for all } x \geq 0.$$

Proposition 4.4.

(a) *If $h_4(0) \geq 0$, then there holds that $h_4(x) > 0$ for all $x > 0$.*
(b) *If $h_4(0) < 0$, then there exists some $b > 0$ such that $h_4(x) < 0 = h_4(b)$ for all $0 \leq x < b$ and $h_4(x) > 0$ for all $x > b$.*

Proof. (a) Since $h''(x) > 0$ for all $x > 0$, we have $h_4' + hh_4 = (h'')^2 > 0$, $x > 0$. Since $h_4(0) \geq 0$, by **Lemma 1**, we have $h_4(x) > 0$ for all $x > 0$.
(b) By the proof of part (a), we only need to show that it's impossible to have that $h_4(x) < 0$ for all $x > 0$. Otherwise, we have $h'''(x) \leq h'''(0)$ for all $x \geq 0$, by integrating this inequality three times, we have $h(x) \leq \frac{1}{6}h'''(0)x^3 + \frac{1}{2}h''(0)x^2 + h'(0)x + h(0)$ for all $x \geq 0$, which implies that $h(x)$ approaches to infinity at most at a cubic polynomial fast rate. This contradicts with the result in **Proposition 4.3**. □

Remark 4.2. By **Theorem 2.7** we know that the maximal solution interval of the equation $h''' + hh'' = (h')^2$ with initial data $h'(0) \geq 0$ and $h''(0) \geq 0$ must be the whole real line $\mathbb{R}$.

(1) When we take the initial data $h(0) = -1$, $h'(0) = 1$ and $h''(0) = 1$, we have $h'''(0) = (h'(0))^2 - h(0)h''(0) = 2 > 0$ and $h^{(4)}(0) = h'(0)h''(0) - h(0)h'''(0) = 3 > 0$. Hence $h'''(x) > 0$ and $h^{(4)}(x) > 0$ for all $x \geq 0$.
(2) When we take the initial data $h(0) = 1$, $h'(0) = 2$ and $h''(0) = 1$, we have $h'''(0) = (h'(0))^2 - h(0)h''(0) = 3 > 0$ and $h^{(4)}(0) = h'(0)h''(0) - h(0)h'''(0) = -1 < 0$. Hence $h'''(x) > 0$ for all $x \geq 0$, and there exists some $b > 0$ such that $h_4(x) < 0 = h_4(b)$ for all $0 \leq x < b$ and $h_4(x) > 0$ for all $x > b$.
(3) When $h'''(0) \leq 0$, that is, $(h'(0))^2 \leq h(0)h''(0)$.
   (i) If $h''(0) = 0$, we obtain $h'(0) = 0$, by the theorem of uniqueness for solutions to the ODE, we must have $h(x) \equiv h(0)$ for all $x \geq 0$.
   (ii) If $h''(0) > 0$, since $(h'(0))^2 \leq h(0)h''(0)$, we obtain $h(0) > 0$. Since $h'(0) \geq 0$, it's easy to see that $h^{(4)}(0) = h'(0)h''(0) - h(0)h'''(0) \geq 0$. Hence $h^{(4)}(x) > 0$ for all $x > 0$.

4.2. *Unbounded solutions when $h'(0) < 0$ and $h''(0) \geq 0$*

In this subsection let's assume that $h(x)$ is an unbounded solution of equation (4.1) with the conditions $h'(0) < 0$ and $h''(0) \geq 0$. We now demonstrate that, under these conditions, the solution $h(x)$ is concave up, possesses a unique critical point, and that for x beyond this point, its behavior matches the pattern established in **Subsection 4.1**.

Proposition 4.5. *There holds that $h''(x) > 0$ for all $x > 0$.*

Proof. The statement follows directly from **Proposition 2.2**. □

Lemma 4.2. *There exists some $x_0 > 0$ such that $h'(x_0) = 0$.*

Proof. If the statement is not true, since $h'(0) < 0$, we must have $h'(x) < 0$ for all $x \geq 0$. Since $h(x)$ is unbounded, we have $\lim_{x \longrightarrow \infty} h(x) = -\infty$. Since $h'(x) < 0$ and $h''(x) > 0$ for all $x > 0$, hence there is some constant $A \leq 0$ such that

$$\lim_{x \longrightarrow \infty} h'(x) = A, \qquad \text{and} \qquad h'(x) < A \quad \text{for all } x \geq 0.$$

Since $h''' = (h')^2 - hh''$ and $h''(x) > 0$ for all $x > 0$, since $\lim_{x \longrightarrow \infty} h(x) = -\infty$, then there exists some large $R_2 \gg 1$ such that $h'''(x) > 0$ for all $x \geq R_2$. Since $h''(x) > 0$ for all $x > 0$, then $\lim_{x \longrightarrow \infty} h''(x) = B$ for some $B \in (0, \infty]$. Since $\lim_{x \longrightarrow \infty} h'(x) = A$, by the mean value theorem, we must have $B = 0$, contradiction. □

Proposition 4.6. *There exists some $c > 0$ such that $h'(x) < 0 = h'(c)$ for all $0 \leq x < c$ and $h'(x) > 0$ for all $x > c$. In particular, since $h'(c) = 0$, $h''(c) > 0$, and $h(x + c)$ is a non-constant solution of (4.1), we know that starting from the unique critical point c, the solution $h(x)$ behaves like the case in **Subsection 4.1** (Fig. 4).*

Proof. The statement follows directly from **Lemma 4.2**, **Proposition 4.5** and **Proposition 4.1**. □

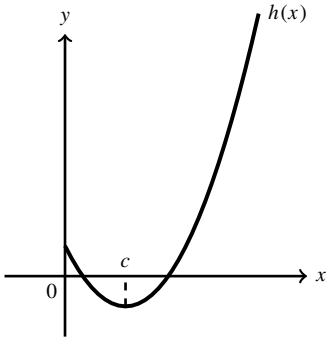


Fig. 4. $h'(0) < 0$ and $h''(0) \geq 0$.

4.3. *Unbounded solutions when $h'(0) \leq 0$ and $h''(0) < 0$*

In this subsection, we assume that $h(x)$ is a solution of equation (4.1) satisfying $h'(0) \leq 0$ and $h''(0) < 0$. Under these conditions, the solution $h(x)$ will be unbounded, and there are two possible types of such unbounded solutions:

(I) $h(x)$ is strictly decreasing, concave down and approaches to negative infinity faster than exponential.
(II) $h(x)$ possesses a unique critical point and a unique inflection point. Beyond its critical point, its behavior mirrors that described in **Subsection 4.1**.

Furthermore, the distinction between these two cases is determined by the interplay of the initial data $h(0)$, $h'(0)$ and $h''(0)$.

Since $h'(0) \leq 0$ and $h''(0) < 0$, it follows that one of two possibilities must hold:

(1) $h'(x) < 0$ for all $x > 0$.
(2) there exists some $c > 0$ such that $h'(x) < 0 = h'(c)$ for all $0 < x < c$.

Proposition 4.7. *Assume that $h'(x) < 0$ for all $x > 0$, then there holds that $h''(x) < 0$ for all $x \geq 0$. In particular, we have $\lim_{x \longrightarrow \infty} h(x) = -\infty$ (Fig. 5).*

Proof. If the statement is not true, since $h''(0) < 0$, then there exists some $a > 0$ such that $h''(x) < 0 = h''(a)$ for all $0 \leq x < a$. Since $h_2' + hh_2 = (h')^2 > 0$ for all $x > 0$, by **Lemma 1**, we obtain $h''(x) > 0$ for all $x > a$. Since $h'(0) \leq 0$ and $h''(0) < 0$, by the analysis for the bounded solutions on $[0, \infty)$ in **Section 3**, we know that $h(x)$ must be unbounded on $[0, \infty)$. Since $h'(x) < 0$ for all $x > 0$, we must have $\lim_{x \longrightarrow \infty} h(x) = -\infty$, which implies that there is some $b > a$ such that $h(x) \leq -1$ for all $x \geq b$. Since $h''(x) > 0$ for all $x > a$, we know that $h_2' = h''' = (h')^2 - hh_2 \geq h_2$ for all $x \geq b$, that is, $(\ln h_2(x))' = \frac{h_2'(x)}{h_2(x)} \geq 1$ for all $x \geq b$. Hence $\ln \frac{h_2(x)}{h_2(b)} \geq x - b$ for all $x \geq b$, that is, $h''(x) = h_2(x) \geq h''(b)e^{x-b}$ for all $x \geq b$. Integrating this inequality twice, we can obtain $h(x) \geq h''(b)(e^{x-b} - x - 1 + b) + h'(b)(x - b) + h(b)$ for all $x \geq b$. Since $h''(b) > 0$, we obtain $\lim_{x \longrightarrow \infty} h(x) = \infty$, which contradicts with the fact that $\lim_{x \longrightarrow \infty} h(x) = -\infty$. □

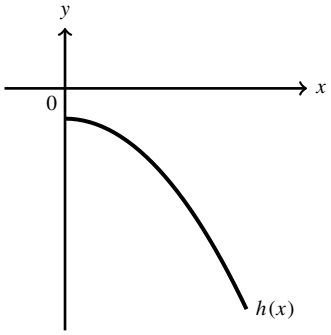


Fig. 5. Type I solution for $h'(0) \le 0$ and $h''(0) < 0$.

Proposition 4.8. *Assume that $h'(x) < 0$ for all $x > 0$, then there holds that $h'''(x) < 0$ for all $x \ge 0$. In particular, $h'''(0) < 0$, that is, $h(0) < \frac{(h'(0))^2}{h''(0)} \le 0$, and $\lim_{x\longrightarrow\infty} h'(x) = -\infty$.*

Proof. If the statement is not true, that is, there exists some $a \ge 0$ such that $h'''(a) \ge 0$. Since $h'(x) < 0$ and $h''(x) < 0$ for all $x > 0$, we have $h_3' + hh_3 = h'h'' > 0$, $x > 0$. By **Lemma 1**, we obtain $h'''(x) > 0$ for all $x > a$. Since $h''(x) < 0$ for all $x \ge 0$, there is some constant $A \le 0$ such that

$$\lim_{x\longrightarrow\infty} h''(x) = A. \tag{4.9}$$

Notice that $h_4 = h'h'' - hh_3$ for all $x \ge 0$. Since $h'(x) < 0$ and $h''(x) < 0$ for all $x > 0$, $\lim_{x\longrightarrow\infty} h(x) = -\infty$, and $h_3(x) > 0$ for all $x > a$, we know that there exists some large $b > a$ such that $h_4(x) > 0$ for all $x \ge b$. Since $h'''(x) > 0$ for all $x > a$, it's easy to have $\lim_{x\longrightarrow\infty} h''(x) = \infty$, which contradicts with the limit (4.9). □

Proposition 4.9. *Assume that $h'(x) < 0$ for all $x > 0$, then there are some constants $b > 0$, $\alpha > 0$, $C > 0$ such that $h(x) \le -Ce^{\alpha x}$ for all $x \ge b$. In particular, $h(x)$ approaches to negative infinity faster than exponential.*

Proof. By **Proposition 4.7** and **Proposition 4.8**, by a right translation, W.L.O.G., we may assume that $h^{(k)}(x) < 0$ for all $x \ge 0$, for all $k = 0, 1, 2, 3$. Let's consider the function $g(x) := -\frac{h''(x)}{h'(x)}$ for all $x \ge 0$, we have

$$g(x) < 0, \qquad g' + hg + h' - g^2 = 0, \quad \text{and} \quad g'' + (h - 2g)g' = 0 \qquad \text{for all } x \ge 0.$$

Since the derivative $g'(x)$ solves the linear equation $y' + (h - 2g)y = 0$, $x > 0$, by the theorem of uniqueness for solutions to the ODE, we have

$$\text{either} \quad g'(x) \equiv 0 \text{ for all } x \ge 0, \quad \text{or} \quad g'(x) > 0 \text{ for all } x \ge 0, \quad \text{or} \quad g'(x) < 0 \text{ for all } x \ge 0. \tag{4.10}$$

Claim I: $g'(x) \le 0$ for all $x \ge 0$.

If the statement is not true, by the alternative in (4.10), we must have $g'(x) > 0$ for all $x \geq 0$. Since $g(x) < 0$ for all $x \geq 0$, then there exists some constant $A \leq 0$ such that $\lim_{x \longrightarrow \infty} g(x) = A$. Since $g'' = -(h - 2g)g'$ and $g'(x) > 0$ for all $x \geq 0$, $\lim_{x \longrightarrow \infty} h(x) = -\infty$, and $\lim_{x \longrightarrow \infty} g(x) = A$, then there exists some large $a > 0$ such that $g''(x) > 0$ for all $x \geq a$. Since $g'(x) > 0$ for all $x \geq 0$, hence $\lim_{x \longrightarrow \infty} g'(x) = B$ some $B \in (0, \infty]$. Since $\lim_{x \longrightarrow \infty} g(x) = A$, by the mean value theorem, we must have $\lim_{x \longrightarrow \infty} g'(x) = B = 0$, which contradicts with the fact $B \in (0, \infty]$.

Since $g'(x) \leq 0$ and $g(x) < 0$ for all $x \geq 0$, we have $-\frac{h''(x)}{h'(x)} = g(x) \leq g(0) < 0$ for all $x \geq 0$. Since $h'(x) < 0$ for all $x \geq 0$, the above inequality is equivalent to $(\ln(-h'(x)))' \geq -g(0) > 0$ for all $x \geq 0$. Integrating this inequality yields that $\ln \frac{-h'(x)}{-h'(0)} \geq -g(0)x$ for all $x \geq 0$, that is, $h'(x) \leq h'(0) \cdot e^{-g(0)x}$ for all $x \geq 0$. Integrating this inequality, we have

$$h(x) \leq -\frac{h'(0)}{g(0)} \cdot (e^{-g(0)x} - 1) + h(0) \qquad \text{for all } x \geq 0.$$

Since $h(x) < 0$ for all $x \geq 0$, $h'(0) < 0$ and $g(0) < 0$, then there are some constants $C > 0$, $\alpha > 0$ such that $h(x) \leq -Ce^{\alpha x}$ for all $x \geq 0$. □

Remark 4.3. When $g'(x) \equiv 0$, $x \geq 0$, in the proof of **Proposition 4.9**, by the same computation as **Remark 4.1**, we know that there are some constants $a < 0$ and $\alpha > 0$ such that

$$h(x) = ae^{\alpha x} - \alpha \qquad \text{for all } x \geq 0.$$

Remark 4.4. In the proof of **Proposition 4.9** we have $g(x) < 0$ and $g'(x) \leq 0$ for all $x \geq 0$, then $\lim_{x \longrightarrow \infty} -g(x) = \lim_{x \longrightarrow \infty} \frac{h''(x)}{h'(x)} = A$ for some $A \in (0, \infty]$. Since $\lim_{x \longrightarrow \infty} h(x) = \lim_{x \longrightarrow \infty} h'(x) = -\infty$, by L'Hospital's rule, we have $\lim_{x \longrightarrow \infty} \frac{\ln(-h(x))}{x} = \lim_{x \longrightarrow \infty} \frac{h'(x)}{h(x)} = \lim_{x \longrightarrow \infty} \frac{h''(x)}{h'(x)} = A$. If $A < \infty$, we would obtain the exact exponential decaying rate of $h(x)$; but if $A = \infty$, $h(x)$ will approach to negative infinity faster than any exponential rate. We expect to have $A < \infty$, which is still open.

Proposition 4.10. *Assume that $h'(x) < 0$ for all $x > 0$, then there holds that $h_4(x) < 0$ for all $x \geq 0$. In particular, we have $h_4(0) < 0$, that is, $(h(0))^2 - \frac{h(0)(h'(0))^2}{h''(0)} + h'(0) > 0$, and $\lim_{x \longrightarrow \infty} h''(x) = -\infty$.*

Proof. If the statement is not true, that is, there is some $a \geq 0$ such that $h_4(a) \geq 0$. Since $h_4' + hh_4 = (h'')^2 > 0$, $x \geq 0$, by **Lemma 1**, we have $h_4(x) > 0$ for all $x > a$. So we have $h'''(x) \geq h'''(a)$ for all $x \geq a$. By integration this inequality three times, we obtain $h(x) \geq \frac{1}{6}h'''(a)(x - a)^3 + \frac{1}{2}h''(a)(x - a)^2 + h'(a)(x - a) + h(a)$ for all $x \geq a$. Since $h'''(a) < 0$, we know that The function $h(x)$ approaches to negative infinity no faster than a cubic polynomial as x approaches to infinity, which contradicts with the result in **Proposition 4.9**.

Differentiating the equation $h''' + hh'' = (h')^2$ yields that $h^{(4)} + h' \cdot h'' + h \cdot h''' = 2h' \cdot h''$, which implies that

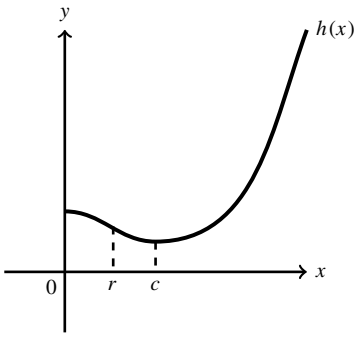


Fig. 6. Type II solution for $h'(0) \le 0$ and $h''(0) < 0$.

$$\begin{aligned} h^{(4)} =& h' \cdot h'' - h \cdot h''' \\ =& h' \cdot h'' - h \cdot ((h')^2 - h \cdot h'') \\ =& h' \cdot h'' - h \cdot (h')^2 + h^2 \cdot h'' \\ =& h'' \cdot \left(h' - \frac{h \cdot (h')^2}{h''} + h^2 \right) \end{aligned}$$

Since $h^{(4)}(0) < 0$ and $h''(0) < 0$, obtain

$$h'(0) - \frac{h(0)(h'(0))^2}{h''(0)} + (h(0))^2 > 0.$$

Since $h^{(4)}(x) < 0$ and $h'''(x) < 0$ for all $x \ge 0$, we have $h'''(x) \le h'''(1) < 0$ for all $x \ge 1$. By integrating this inequality, we have $h''(x) \le h'''(1) \cdot (x-1)$ for all $x \ge 1$. Since $h'''(1) < 0$, then $\lim_{x \longrightarrow \infty} h''(x) = -\infty$. □

Proposition 4.11. *Assume that there exists some $c > 0$ such that $h'(x) < 0 = h'(c)$ for all $0 < x < c$, then there exists some $r \in (0, c)$ such that*

$$\begin{aligned} &h''(x) < 0 = h''(r) \quad \text{for all } 0 \le x < r, \qquad \text{and} \qquad h''(x) > 0 \quad \text{for all } x > r \\ &h'(x) < 0 = h'(c) \quad \text{for all } 0 < x < c, \qquad \text{and} \qquad h'(x) > 0 \quad \text{for all } x > c. \end{aligned}$$

*In particular, since $h'(c) = 0$ and $h''(c) > 0$, we know that after the critical point c, the solution $h(x)$ behaves like the case in **Subsection 4.1** (Fig. 6).*

Proof. Since $h'(x) < 0 = h'(c)$ for all $0 < x < c$, it's easy to see that $h''(c) \ge 0$. Since $h_2' + hh_2 = (h')^2 \ge 0$, $x \ge 0$, and $h_2(c) = h''(c) \ge 0$, by **Lemma 1**, we have $h''(x) \ge 0$ for all $x \ge c$. Since $h_2'' + hh_2' - h'h_2 = 0$, $x \ge 0$, and $h_2(x) = h''(x) \ge 0$ for all $x \ge c$, by the strong maximum principle, we have

$$\text{either} \qquad h''(x) = h_2(x) \equiv 0 \text{ for all } x \ge c, \qquad \text{or} \qquad h''(x) = h_2(x) > 0 \text{ for all } x > c.$$

Since $h''(0) < 0$, and only linear solutions to the equation (4.1) are constant functions, we must have $h''(x) > 0$ for all $x > c$. Since $h''(0) < 0$, there exists some $r \in (0, c]$ such that

$$h''(x) < 0 = h''(r) \quad \text{for all } 0 \le x < r.$$

By the same argument as the beginning, we can obtain that

$$h''(x) > 0 \qquad \text{for all } x > r.$$

Since $h'(0) \le 0$ and $h''(x) < 0$ for all $0 \le x < r$, it's easy to know that $h'(r) < 0$, which implies that

$$r < c.$$

Since $c > r$, $h'(c) = 0$, and $h''(x) > 0$ for all $x > r$, we obtain

$$h'(x) > 0 = h'(c) \qquad \text{for all } x > c. \quad \square$$

Corollary 4.2. *If $h'''(0) \ge 0$, that is, $h(0) \ge \frac{(h'(0))^2}{h''(0)}$, then there are some constants $0 < r < c$ such that*

$$\begin{aligned} &h''(x) < 0 = h''(r) \quad \text{for all } 0 \le x < r, \qquad \text{and} \qquad h''(x) > 0 \quad \text{for all } x > r \\ &h'(x) < 0 = h'(c) \quad \text{for all } 0 < x < c, \qquad \text{and} \qquad h'(x) > 0 \quad \text{for all } x > c. \end{aligned}$$

Proof. The statement follows directly from **Proposition 4.8** and **Proposition 4.11**. $\square$

Corollary 4.3. *If $h^{(4)}(0) \ge 0$, that is, $h'(0) - \frac{h(0)(h'(0))^2}{h''(0)} + (h(0))^2 \le 0$, then there are some constants $0 < r < c$ such that*

$$\begin{aligned} &h''(x) < 0 = h''(r) \quad \text{for all } 0 \le x < r, \qquad \text{and} \qquad h''(x) > 0 \quad \text{for all } x > r \\ &h'(x) < 0 = h'(c) \quad \text{for all } 0 < x < c, \qquad \text{and} \qquad h'(x) > 0 \quad \text{for all } x > c. \end{aligned}$$

Proof. The statement follows directly from **Proposition 4.10** and **Proposition 4.11**. $\square$

Corollary 4.4. *Assume that $h'(0) = 0$, and $h''(0) < 0$, then there are some constants $0 < r < c$ such that*

$$\begin{aligned} &h''(x) < 0 = h''(r) \quad \text{for all } 0 \le x < r, \qquad \text{and} \qquad h''(x) > 0 \quad \text{for all } x > r \\ &h'(x) < 0 = h'(c) \quad \text{for all } 0 < x < c, \qquad \text{and} \qquad h'(x) > 0 \quad \text{for all } x > c. \end{aligned}$$

Proof. If $h(0) \ge 0$, we have $h'''(0) = (h'(0))^2 - h(0)h''(0) = -h(0)h''(0) \ge 0$, the statement follows directly from **Corollary 4.2**. In the following let's assume that $h(0) < 0$, if the statement is not true, since $h'(0) = 0$ and $h''(0) < 0$, by **Proposition 4.11**, we must have $h'(x) < 0$ for all $x > 0$. Since $h(0) < 0$, by **Proposition 4.7** and **Proposition 4.8**, we have

$$h^{(k)}(x) < 0 \quad \text{for all } x > 0, \qquad \text{for all } k = 0, 1, 2, 3, \qquad \text{and} \qquad \lim_{x \longrightarrow \infty} h(x) = -\infty.$$

Let's consider the function $g(x) := -\frac{h''(x)}{h'(x)}$ for all $x > 0$, we have

$$g(x) < 0, \qquad g' + hg + h' - g^2 = 0, \quad \text{and} \quad g'' + (h - 2g)g' = 0 \qquad \text{for all } x > 0.$$

Since the derivative $g'(x)$ solves $y' + (h - 2g)y = 0$, $x > 0$, by the theorem of uniqueness for solutions to the ODE, we have

$$\text{either} \quad g'(x) \equiv 0 \text{ for all } x \geq 0, \quad \text{or} \quad g'(x) > 0 \text{ for all } x \geq 0, \quad \text{or} \quad g'(x) < 0 \text{ for all } x > 0. \tag{4.11}$$

Claim I: $g'(x) \leq 0$ for all $x > 0$.

If the statement is not true, by the alternative in (4.11), we should have $g'(x) > 0$ for all $x > 0$. Since $g(x) < 0$ for all $x > 0$, then there exists some constant $A \leq 0$ such that

$$\lim_{x \longrightarrow \infty} g(x) = A, \qquad \text{and} \qquad g(x) < A \quad \text{for all } x > 0.$$

On the other hand, since $g'' = -(h - 2g)g'$ and $g'(x) > 0$ for all $x > 0$, $\lim_{x \longrightarrow \infty} h(x) = -\infty$, and $\lim_{x \longrightarrow \infty} g(x) = A$, then there exists some large $a > 0$ such that $g''(x) > 0$ for all $x \geq a$. Since $g'(x) > 0$ for all $x \geq 0$, hence $\lim_{x \longrightarrow \infty} g'(x) = B$ for some $B \in (0, \infty]$. Since $\lim_{x \longrightarrow \infty} g(x) = A$, by the mean value theorem, we have $\lim_{x \longrightarrow \infty} g'(x) = B = 0$, contradiction.

Claim II: $g'(x) \equiv 0$ for all $x > 0$.

If the statement is not true, by **Claim I** and the alternative in (4.11), we should have $g'(x) < 0$ for all $x > 0$. So we have $-\frac{h''(x)}{h'(x)} = g(x) \geq g(1)$ for all $0 < x \leq 1$, which implies that

$$\liminf_{x \longrightarrow 0^+} -\frac{h''(x)}{h'(x)} \geq g(1). \tag{4.12}$$

On the other hand, since $h''(0) < 0$, $h'(0) = 0$, and $h'(x) < 0$ for all $x > 0$, we know that $\lim_{x \longrightarrow 0^+} -\frac{h''(x)}{h'(x)} = -\infty$, which contradicts with the lower bound estimate in (4.12).

By **Claim II** and the computations in **Remark 4.1**, we know that there are some constants $\alpha > 0$ and $a < 0$ such that $h(x) = ae^{\alpha x} - \alpha$, $x > 0$. By the continuity, we have $h(x) = ae^{\alpha x} - \alpha$, $x \geq 0$. This gives us that $h'(0) = a\alpha < 0$, which contradicts with the assumption that $h'(0) = 0$. □

4.4. Unbounded solutions when $h'(0) > 0$ and $h''(0) < 0$

In this subsection, we consider an unbounded solution $h(x)$ of equation (4.1) satisfying $h'(0) > 0$ and $h''(0) < 0$. There are two possible types of such solutions. The first type is strictly increasing, possesses a unique inflection point, and beyond this point, its behavior mirrors that described in **Subsection 4.1**. The second type has two critical points; after the first critical point, its behavior aligns with the second case presented in **Subsection 4.3**.

Since $h'(0) > 0$ and $h''(0) < 0$, we know that

$$\begin{aligned} &\text{either } h'(x) > 0 \text{ for all } x \geq 0, \\ &\quad \text{or} \quad \text{there exists some } c > 0 \text{ such that } h'(x) > 0 = h'(c) \text{ for all } 0 \leq x < c. \end{aligned}$$

Lemma 4.3. *Assume that $h'(x) > 0$ for all $x \geq 0$, then there exists some $x_0 > 0$ such that $h''(x_0) = 0$.*

Proof. If the statement is not true, since $h''(0) < 0$, we should have $h''(x) < 0$ for all $x \geq 0$. Since $h'(x) > 0$ for all $x \geq 0$, there is some constant $A \geq 0$ such that

$$\lim_{x \longrightarrow \infty} h'(x) = A.$$

Since $h'(x) > 0$ for all $x \geq 0$ and h is unbounded on $[0, \infty)$, we must have $\lim_{x \longrightarrow \infty} h(x) = \infty$. Since $h''' = (h')^2 - hh''$ and $h''(x) < 0$ for all $x \geq 0$, it's easy to know that there exists some large $R_1 > 0$ such that $h'''(x) > 0$ for all $x \geq R_1$. Since $h_3' + hh_3 = h'h'' < 0$, $x \geq 0$, by **Lemma 1**, we must have $h'''(x) > 0$ for all $x \geq 0$. Since $\lim_{x \longrightarrow \infty} h'(x) = A$, by the mean value theorem, we have

$$\lim_{x \longrightarrow \infty} h''(x) = 0.$$

Claim I: $h_4(x) < 0$ for all $x \geq 0$.

If the assertion is not true, that is, there exists some $R_2 \geq 0$ such that $h_4(R_2) \geq 0$. Since $h_4' + hh_4 = (h'')^2 > 0$, $x \geq 0$, by **Lemma 1**, we have $h_4(x) > 0$ for all $x > R_2$. Since $h'''(x) > 0$ for all $x \geq 0$, we have $h'''(x) \geq h'''(R_2) =: \alpha > 0$ for all $x \geq R_2$. Integrating this inequality three times, we know that there are some constants $\beta > 0$ and $R_3 > R_2$ such that

$$h(x) \geq \beta x^3 \qquad \text{for all } x \geq R_3. \tag{4.13}$$

On the other hand, since $h''(x) < 0$ for all $x \geq 0$, we have $h'(x) \leq h'(0)$ for all $x \geq 0$, which implies that $h(x) = h(0) + \int_0^x h'(t)\, dt \leq h(0) + \int_0^x h'(0)\, dt = h'(0)x + h(0)$ for all $x \geq 0$. This contradicts with the lower bound estimate (4.13).

Since $\lim_{x \longrightarrow \infty} h''(x) = 0$, by the mean value theorem and **Claim I**, we have $\lim_{x \longrightarrow \infty} h'''(x) = 0$. Since $h''' + hh'' = (h')^2$ and $h''(x) < 0$ for all $x \geq 0$, $\lim_{x \longrightarrow \infty} h(x) = \infty$, and $\lim_{x \longrightarrow \infty} h'(x) = A \geq 0$, we must have

$$\lim_{x \longrightarrow \infty} h(x)h''(x) = \lim_{x \longrightarrow \infty} h'(x) = A = 0.$$

Let's define the function $f(x) := h_2(x) - \frac{h_2(0)}{h_1(0)} h_1(x)$, $x \geq 0$, by the equations (2) and (3) with $\beta = 0$, we know that $f'' + hf' - h'f = 0$, $x \geq 0$. Since $f(0) = 0$, $h_1(\infty) = h_2(\infty) = 0$ and $-h'(x) < 0$ for all $x \geq 0$, by the strong maximum principle, we have $f(x) \equiv 0$ for all $x \geq 0$. Let $\alpha := -\frac{h''(0)}{h'(0)} > 0$, we have $h'' = -\alpha h'$, $x \geq 0$, which implies that there are some $A, B \in \mathbb{R}$ such that $h = A + Be^{-\alpha x}$, $x \geq 0$, which is a bounded function on $[0, \infty)$, contradiction. □

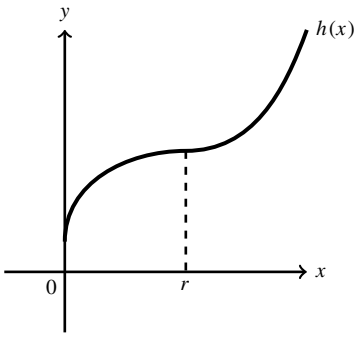


Fig. 7. Type I solution for $h'(0) > 0$ and $h''(0) < 0$.

Proposition 4.12. *Assume that $h'(x) > 0$ for all $x \geq 0$, then there exists some $r > 0$ such that $h''(x) < 0 = h''(r)$ for all $0 \leq x < r$ and $h''(x) > 0$ for all $x > r$. In particular, since $h'(r) > 0$, $h''(r) = 0$, and $h(x + r)$ is a non-constant solution of (4.1), we know that after the unique reflection point r, the solution $h(x)$ behaves like the case in **Subsection 4.1** (Fig. 7).*

Proof. Since $h''(0) < 0$, by **Lemma 4.3**, there is some $r > 0$ such that $h''(x) < 0 = h''(r)$ for all $0 \leq x < r$. Since $h_2' + hh_2 = (h')^2 > 0$, $x \geq 0$, by **Lemma 1**, we have $h''(x) > 0$ for all $x > r$. □

Example 4.1. Let $f(x)$ be the solution of the equation $h''' + hh'' = (h')^2$ with the initial data $h(0) = 0$, $h'(0) = 1$ and $h''(0) = 0$, by **Theorem 2.7**, we know that the maximal solution interval of $f(x)$ is $\mathbb{R}$. Since $f'''(0) = (f'(0))^2 - f(0)f''(0) = 1 > 0$, according to the analysis in **Subsection 4.1**, we know that

$$f'(x) > 0 \quad \text{and} \quad f'''(x) > 0 \qquad \text{for all } x \geq 0.$$

Since $f'(0) = 1 > 0$, $f''(0) = 0$, and $f'''(0) = 1 > 0$, so there is some constant $a < 0$ such that

$$f'(x) > 0, \quad \text{and} \quad f''(x) < 0 \qquad \text{for all } a \leq x \leq 0.$$

Let $h(x) := f(x + a)$ for all $x \geq 0$, it's easy to see that $h(x)$ is a solution of equation (4.1) with $h'(0) = f'(a) > 0$, $h''(0) = f''(a) < 0$, and $h'(x) = f'(x + a) > 0$ for all $x \geq 0$.

Lemma 4.4. *Assume that there is some $c > 0$ such that $h'(x) > 0 = h'(c)$ for all $0 \leq x < c$, then we have $h''(x) < 0$ for all $0 \leq x < c$.*

Proof. If the statement is not true, since $h''(0) < 0$, so there is some $0 < x_0 \leq c$ such that $h''(x) < 0 = h''(x_0)$ for all $0 \leq x < x_0$. Since $h'(c) = 0$ and $h(x)$ is not a constant function on $[c, \infty)$, by the theorem of uniqueness for solutions to the ODE, we know that

$$0 < x_0 < c.$$

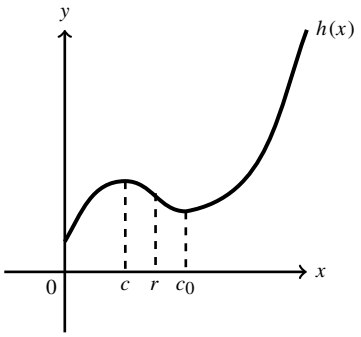


Fig. 8. Type II solution for $h'(0) > 0$ and $h''(0) < 0$.

Since $h_2' + hh_2 = (h')^2 \geq 0$ for all $x \geq 0$, by **Lemma 1**, we have $h''(x) \geq 0$ for all $x \geq x_0$. Since $h_2'' + hh_2' - h'h_2 = 0$ for all $x \geq 0$, by the strong maximum principle, we have

$$\text{either} \qquad h''(x) \equiv 0 \text{ for all } x \geq x_0, \qquad \text{or} \qquad h''(x) > 0 \text{ for all } x > x_0.$$

Since only linear solutions of equation (4.1) are constant functions, so we must have $h''(x) > 0$ for all $x > x_0$. Since $0 < x_0 < c$, we have $h''(c) > 0$. But since $h'(x) > 0 = h'(c)$ for all $0 \leq x < c$, we obtain $h''(c) \leq 0$, we get a contradiction. □

Proposition 4.13. *Assume that there is some $c > 0$ such that $h'(x) > 0 = h'(c)$ for all $0 \leq x < c$, then we have $h''(c) < 0$. In particular, since $h'(c) = 0$, $h''(c) < 0$, and $h(x + c)$ is a non-constant solution of (4.1), by **Corollary 4.4**, we know that beyond the critical point c, the solution $h(x)$ behaves like the second case in **Subsection 4.3** (Fig. 8).*

Proof. If the assertion is not true, we have $h''(c) \geq 0$. By **Lemma 4.4**, we have $h''(c) \leq 0$. Hence $h''(c) = 0$. Since $h'(c) = 0$, by the theorem of uniqueness for solutions to the ODE, we know that $h(x) \equiv h(0)$ for all $x \geq 0$, which contradicts with the assumption that $h'(x) > 0$ for all $0 \leq x < c$. □

Example 4.2. Let $f(x)$ be the solution of the equation $h''' + hh'' = (h')^2$ with the initial data $h(0) = 1$, $h'(0) = 0$ and $h''(0) = -1$, by **Theorem 2.7**, we know that the maximal solution interval of f is $\mathbb{R}$. Since $f'''(0) = (f'(0))^2 - f(0)f''(0) = 1 > 0$, according to the analysis for the bounded solutions in **Section 3**, and the analysis in **Subsection 4.3**, we know that h is unbounded on $[0, \infty)$ and there exist some $0 < r < c$ such that

$$f''(x) < 0 = f''(r) \quad \text{for all } 0 \leq x < r, \qquad \text{and} \qquad f''(x) > 0 \quad \text{for all } x > r$$
$$f'(x) < 0 = f'(c) \quad \text{for all } 0 < x < c, \qquad \text{and} \qquad f'(x) > 0 \quad \text{for all } x > c.$$

Since $f'(0) = 0$, $f''(0) = -1$, and $f'''(0) = 1 > 0$, so there is some constant $a < 0$ such that

$$f'(x) > 0, \quad \text{and} \quad f''(x) < 0 \qquad \text{for all } a \leq x < 0.$$

Let $h(x) := f(x + a)$ for all $x \geq 0$, it's easy to see that $h(x)$ is a solution of the equation (4.1) with $h''(0) < 0$ and $h'(x) > 0 = h'(-a)$ for all $0 \leq x < -a$.

4.5. Unbounded solutions on the left line

For the unbounded solution to the equation $h''' + hh'' = (h')^2$, $x \le 0$, notice that the function $H(x) := -h(-x)$, $x \ge 0$, is an unbounded solution to the equation $h''' + hh'' = (h')^2$, $x \ge 0$, applying earlier results, we obtain the following classification:

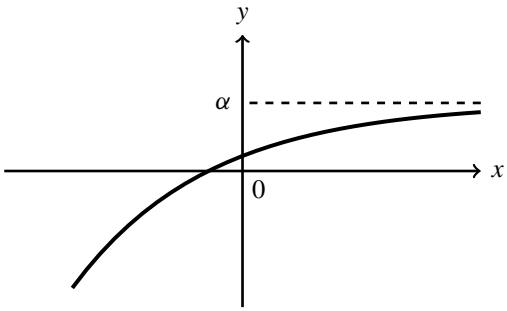
Proposition 4.14. *Let $h(x)$ be an unbounded solution to the equation $h''' + hh'' = (h')^2$, $x \le 0$.*

(a) *If $h'(0) \ge 0$ and $h''(0) \le 0$, then $h(x)$ is increasing and concave down on $(-\infty, 0)$, and $h(x)$ approaches to negative infinity at an exponential rate as x goes to negative infinity.*
(b) *If $h'(0) < 0$ and $h''(0) \le 0$, then $h(x)$ is concave down on $(-\infty, 0)$ and has a unique critical point $c < 0$, on the left hand side of the critical point c, $h(x)$ behaves like the case in part (a).*
(c) *If $h'(0) \le 0$ and $h''(0) > 0$, then either $h'(x) < 0$ and $h''(x) > 0$ for all $x < 0$, and $h(x)$ approaches to infinity faster than an exponential rate as x goes to negative infinity, or $h(x)$ has a unique critical point $c < 0$ and a unique reflection point $r \in (c, 0)$, $h(x)$ is decreasing and concave up on $(r, 0)$, and on the left hand side of the reflection point r, $h(x)$ behaves like the case in the part (b).*
(d) *If $h'(0) > 0$ and $h''(0) > 0$, then either $h(x)$ is increasing, has a unique reflection point r, and on the left hand side of the reflection point r, $h(x)$ behaves like the case in the part (a); or $h(x)$ has two critical points $c_1 < c_2 < 0$, and on the left hand side of the second critical point c_2, $h(x)$ behaves like the second case in the part (c).*

5. Proof of Theorem 1.1

Proof of Theorem 1.1. W.L.O.G., let's assume that h is a non-constant solution to equation (1.13).
Case I: If $h'(0) \ge 0$ and $h''(0) \le 0$. By **Theorem 2.7**, we know that the maximal solution interval of $h(x)$ is $\mathbb{R}$.

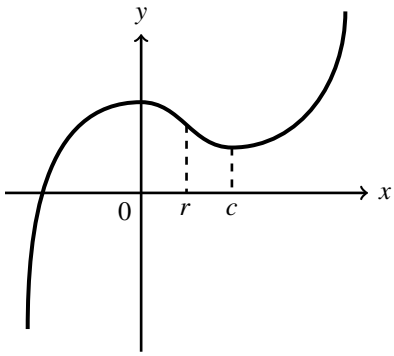
SubCase I: If $h(x)$ is bounded on $[0, \infty)$. By **Theorem 3.1** and **Theorem 3.2**, there are some constants $a < 0$ and $\alpha < 0$ such that $h(x) = ae^{\alpha x} - \alpha$, $x \ge 0$. By the uniqueness theorem for ODEs, then $h(x) = ae^{\alpha x} - \alpha$ for all $x \in \mathbb{R}$.



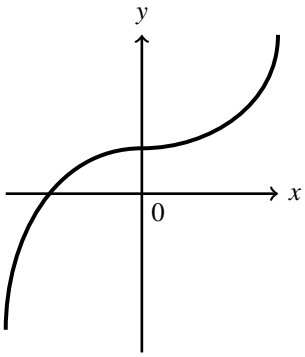
SubCase II: If $h(x)$ is bounded on $(-\infty, 0]$. By **Theorem 3.3**, it is impossible to have $h'(0) \ge 0$ and $h''(0) \le 0$.

SubCase III: If $h(x)$ is both unbounded on $[0, \infty)$ and $(-\infty, 0]$. If $h'(0) = 0$, since $h''(0) \le 0$, by the uniqueness theorem for ODEs, we have $h''(0) < 0$. By the analysis in **Subsection 4.3**

and **Proposition 4.14**, there are some constants $0 < r < c$ such that $h'(x) < 0 = h'(c)$ for all $0 < x < c$, $h'(x) > 0$ for all $x < 0$ or $x > c$, $h''(x) < 0 = h''(r)$ for all $x < r$, $h''(x) > 0$ for all $x > r$, $h(x)$ approaches to infinity at an exponential rate as $x \longrightarrow +\infty$, and $h(x)$ approaches to negative infinity at an exponential rate as $x \longrightarrow -\infty$.

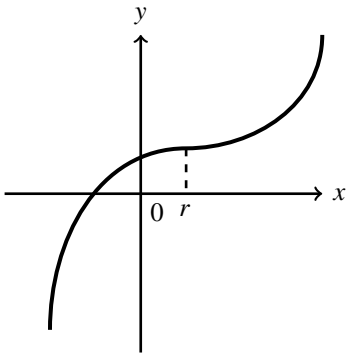


If $h'(0) > 0$ and $h''(0) = 0$, by the analysis in **Subsection 4.1** and **Proposition 4.14**, then $h'(x) > 0$ for all $x \in \mathbb{R}$, $h''(x) < 0$ for all $x < 0$, $h''(x) > 0$ for all $x > 0$, $h(x)$ approaches to infinity at an exponential rate as $x \longrightarrow +\infty$, and $h(x)$ approaches to negative infinity at an exponential rate as $x \longrightarrow -\infty$.

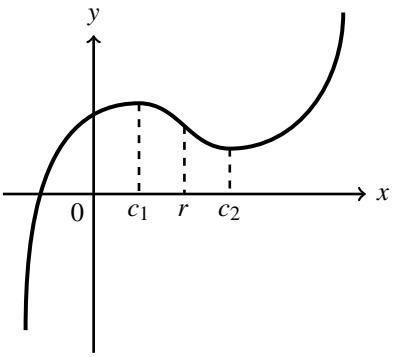


If $h'(0) > 0$ and $h''(0) < 0$, by the analysis in **Subsection 4.4** and **Proposition 4.14**, we know that

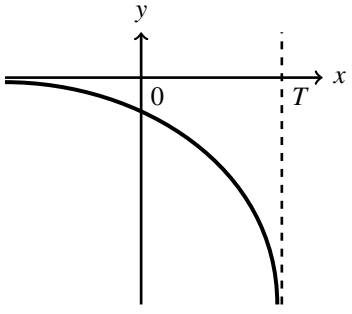
either (1) $h'(x) > 0$ for all $x \in \mathbb{R}$, there exists some constant $r > 0$ such that $h''(x) < 0 = h''(r)$ for all $x < r$, $h''(x) > 0$ for all $x > r$, $h(x)$ approaches to infinity at an exponential rate as $x \longrightarrow +\infty$, and $h(x)$ approaches to negative infinity at an exponential rate as $x \longrightarrow -\infty$.



or (2) there are some $0 < c_1 < r < c_2$ such that $h'(x) < 0 = h'(c_1) = h'(c_2)$ for all $c_1 < x < c_2$, $h'(x) > 0$ for all $x < c_1$ or $x > c_2$, $h''(x) < 0$ for all $x < r$, $h''(x) > 0 = h''(r)$ for all $x > r$, $h(x)$ approaches to infinity at an exponential rate as $x \longrightarrow +\infty$, and $h(x)$ approaches to negative infinity at an exponential rate as $x \longrightarrow -\infty$.



Case II: If $h'(0) < 0$ and $h''(0) < 0$. By **Theorem 2.7**, we know that the maximal solution interval is either $\mathbb{R}$ or $(-\infty, T)$ for some $0 < T < \infty$. If $0 < T < \infty$, we must have $h^{(k)}(x) < 0$, $x < T$, for each $k = 0, 1, 2, 3, 4$, and $\lim_{x \longrightarrow T^-} h(x) = -\infty$, which implies that $h(x)$ is a bounded function on $(-\infty, 0]$. By **Theorem 3.3** and the uniqueness theorem for ODEs, we know that $\lim_{x \longrightarrow -\infty} h(x) = 0$.

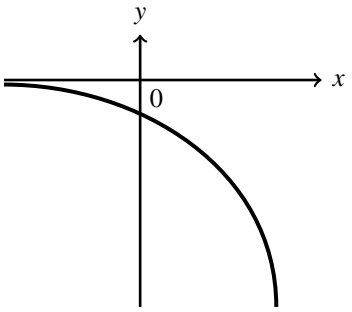


In the following let us assume that $T = \infty$.

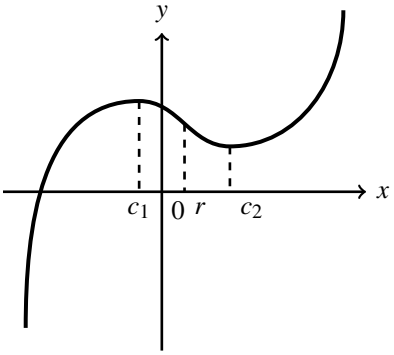
SubCase I: If $h(x)$ is bounded on $(-\infty, 0]$. By **Theorem 3.3**, then $h'(x) < 0$ for all $x \leq 0$, and the limit $\alpha := \lim_{x \longrightarrow -\infty} -h(x)$ exists for some constant $\alpha \geq 0$.

If $\alpha > 0$, by **Theorem 3.3**, then there exists constant $a < 0$ such that $h(x) = ae^{\alpha x} - \alpha$, $x \leq 0$. By the uniqueness theorem for ODEs, then $h(x) = ae^{\alpha x} - \alpha$ for all $x \in \mathbb{R}$.

If $\alpha = 0$, by **Theorem 3.3**, then $h^{(k)}(x) < 0$, $x \leq 0$, for all $k = 0, 1, 2, 3$. Since $\lim_{x \longrightarrow -\infty} h(x) = 0$, by a left shifting of $h(x)$, applying **Theorem 3.3** again, we know that $h^{(k)}(x) < 0$, $x \in \mathbb{R}$, for all $k = 0, 1, 2, 3$. Since $h'(0) < 0$ and $h''(0) < 0$, by **Theorem 3.1**, we know that $h(x)$ is unbounded on $[0, \infty)$. By the analysis in **Subsection 4.3**, we know that $h(x)$ approaches to negative infinity faster than an exponential rate as $x \longrightarrow \infty$.

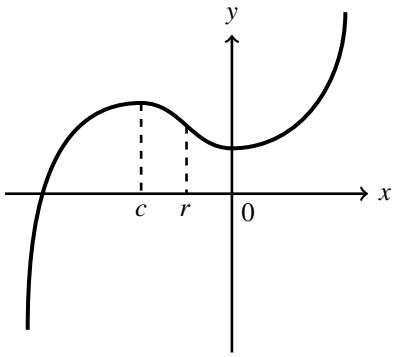


SubCase II: If $h(x)$ is unbounded on $(-\infty, 0]$. By **Proposition 4.14**, then there exists some $c_1 < 0$ such that $h'(x) < 0 = h'(c_1)$ for all $c_1 < x \leq 0$, $h'(x) > 0$ for all $x < c_1$, $h''(x) < 0$ for all $x \leq 0$, and $h(x)$ approaches to negative infinity at an exponential rate as $x \longrightarrow -\infty$. Since $h'(c_1) = 0$ and $h''(c_1) < 0$, since $h(x + c_1)$ is a solution of the equation (1.13) on $[0, \infty)$. By the analysis in **Subsection 3.1** about bounded solutions, we know that $h(x + c_1)$ is unbounded on $[0, \infty)$. By **Corollary 4.4**, then there are some constants $0 < r < c_2$ such that $h'(x) < 0 = h'(c_2)$ for all $0 \leq x < c_2$, $f'(x) > 0$ for all $x > c_2$, $h''(x) < 0 = h''(r)$ for all $0 \leq x < r$, $h''(x) > 0$ for all $x > r$, and $h(x)$ approaches to infinity at an exponential rate as $x \longrightarrow \infty$.



Case III: If $h'(0) = 0$ and $h''(0) > 0$. Let $H(x) := -h(-x)$, so $H(x)$ is a solution of the equation (1.13) with $H'(0) = h'(0) < 0$ and $H''(0) = -h''(0) < 0$. By the result of **Case I**, we know that the maximal solution interval of $H(x)$ is $\mathbb{R}$, and there are some constants $0 < -r < -c$ such that $H'(x) < 0 = H'(-c)$ for all $0 < x < -c$, $H'(x) > 0$ for all $x < 0$ or $x > -c$, $H''(x) < 0 =$

$H''(-r)$ for all $x < -r$, $H''(x) > 0$ for all $x > -r$, $H(x)$ approaches to infinity at an exponential rate as $x \longrightarrow +\infty$, and $H(x)$ approaches to negative infinity at an exponential rate as $x \longrightarrow -\infty$. Since $H(x) = -h(-x)$, then the maximal solution interval of $h(x)$ is $\mathbb{R}$, $h'(x) < 0 = h'(c)$ for all $c < x < 0$, $h'(x) > 0$ for all $x > 0$ or $x < c$, $h''(x) > 0 = h''(r)$ for all $x > r$, $h''(x) < 0$ for all $x < r$, $h(x)$ approaches to negative infinity at an exponential rate as $x \longrightarrow -\infty$, and $h(x)$ approaches to infinity at an exponential rate as $x \longrightarrow \infty$.



Case IV: If $h'(0) < 0$ and $h''(0) > 0$. Let $H(x) := -h(-x)$, so $H(x)$ is a solution of the equation (1.13) with $H'(0) = h'(0) < 0$ and $H''(0) = -h''(0) < 0$. By the results of **Case II**, we know that

either (1) the maximal solution interval of $H(x)$ is $(-\infty, T)$ for some $0 < T < \infty$, $H^{(k)}(x) < 0$, $x < T$, for each $k = 0, 1, 2, 3, 4$, and $\lim_{x \longrightarrow T^-} H(x) = -\infty$, and $\lim_{x \longrightarrow -\infty} H(x) = 0$.

or (2) there exists some constants $-a < 0$ and $-\alpha > 0$ such that $H(x) = -ae^{-\alpha x} + \alpha$ for all $x \in \mathbb{R}$.

or (3) the maximal solution interval of $H(x)$ is $\mathbb{R}$, $H^{(k)}(x) < 0$, $x \in \mathbb{R}$, for each $k = 0, 1, 2, 3$, $\lim_{x \longrightarrow -\infty} H(x) = 0$, and $H(x)$ approaches to negative infinity faster than an exponential rate as $x \longrightarrow \infty$.

or (4) the maximal solution interval of $H(x)$ is $\mathbb{R}$, there are some constants $-c_2 < 0 < -r < -c_1$ such that $H'(x) < 0 = H'(-c_1) = H'(-c_2)$ for all $-c_2 < x < -c_1$, $H'(x) > 0$ for all $x < -c_2$ or $x > -c_1$, $H''(x) < 0$ for all $x < -r$, $H''(x) > 0$ for all $x > -r$, $H(x)$ approaches to infinity at an exponential rate as $x \longrightarrow \infty$, and $H(x)$ approaches to negative infinity at an exponential rate as $x \longrightarrow -\infty$.

Since $H(x) = -h(-x)$, we obtain that

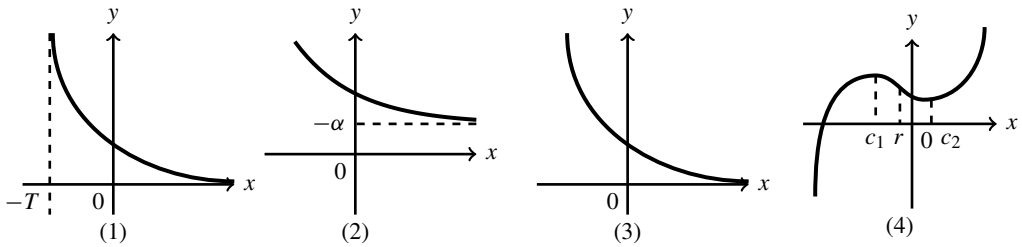
either (1) the maximal solution interval of $h(x)$ is $(-T, \infty)$ for some $0 < T < \infty$, $(-1)^{k+1} h^{(k)}(x) < 0$ for all $x > -T$, for each $k = 0, 1, 2, 3, 4$, and $\lim_{x \longrightarrow -T^+} h(x) = \infty$, and $\lim_{x \longrightarrow \infty} h(x) = 0$.

or (2) there exists some constants $a > 0$ and $\alpha < 0$ such that $h(x) = ae^{\alpha x} - \alpha$, $x \in \mathbb{R}$.

or (3) the maximal solution interval of $h(x)$ is $\mathbb{R}$, $(-1)^{k+1} h^{(k)}(x) < 0$, $x \in \mathbb{R}$, for each $k = 0, 1, 2, 3$, $\lim_{x \longrightarrow \infty} h(x) = 0$, and $h(x)$ approaches to infinity faster than an exponential rate as $x \longrightarrow -\infty$.

or (4) the maximal solution interval of $H(x)$ is $\mathbb{R}$, there are some constants $c_1 < r < 0 < c_2$ such that $h'(x) < 0 = h'(c_1) = h'(c_2)$ for all $c_1 < x < c_2$, $h'(x) > 0$ for all $x > c_2$ or $x < c_1$, $h''(x) < 0$ for all $x < r$, $h''(x) > 0$ for all $x > r$, $h(x)$ approaches to negative infinity at an

exponential rate as $x \longrightarrow -\infty$, and $h(x)$ approaches to infinity at an exponential rate as $x \longrightarrow \infty$.

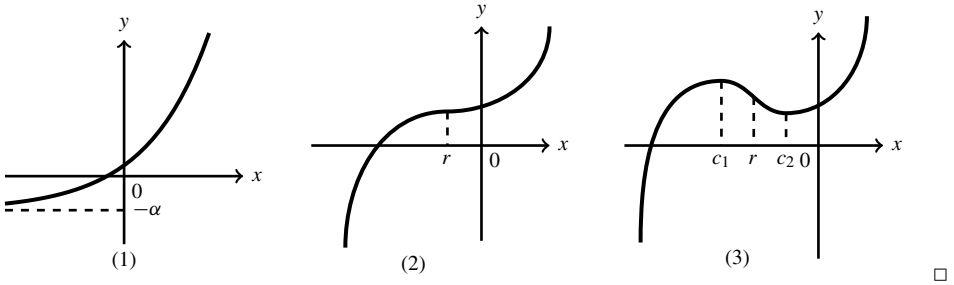


Case V: If $h'(0) > 0$ and $h''(0) > 0$. By **Theorem 2.7**, we know that the maximal solution interval of $h(x)$ is $\mathbb{R}$. Let $H(x) := -h(-x)$, so $H(x)$ is a solution of the equation (1.13) with $H'(0) = h'(0) > 0$ and $H''(0) = -h''(0) < 0$. By the results of **Case I**, we know that

either (1) there are some constants $-a < 0$ and $-\alpha < 0$ such that $H(x) = -ae^{-\alpha x} + \alpha$, $x \geq 0$.
or (2) $H'(x) > 0$ for all $x \in \mathbb{R}$, there exists some $-r > 0$ such that $H''(x) > 0 = H''(-r)$ for all $x > -r$, $H''(x) < 0$ for all $x < -r$, $H(x)$ approaches to infinity at an exponential rate as $x \longrightarrow \infty$, and $h(x)$ approaches to negative infinity at an exponential rate as $x \longrightarrow -\infty$.
or (3) there are some constants $0 < -c_2 < -r < -c_1$ such that $H'(x) < 0 = H'(-c_1) = H'(-c_2)$ for all $-c_2 < x < -c_1$, $H'(x) > 0$ for all $x < -c_2$ or $x > -c_1$, $H''(x) > 0 = H''(-r)$ for all $x > -r$, $H''(x) < 0$ for all $x < -r$, $H(x)$ approaches to infinity at an exponential rate as x goes to infinity, and $H(x)$ approaches to negative infinity at an exponential rate as x goes to negative infinity.

Since $H(x) = -h(-x)$, we obtain that

either (1) there are some constants $a > 0$ and $\alpha > 0$ such that $h(x) = ae^{\alpha x} - \alpha$, $x \in \mathbb{R}$.
or (2) $h'(x) > 0$ for all $x \in \mathbb{R}$, there exists some $r < 0$ such that $h''(x) < 0 = h''(r)$ for all $x < r$, $h''(x) > 0$ for all $x > r$, $h(x)$ approaches to negative infinity at an exponential rate as $x \longrightarrow -\infty$, and $h(x)$ approaches to infinity at an exponential rate as $x \longrightarrow \infty$.
or (3) there are some constants $c_1 < r < c_2 < 0$ such that $h'(x) < 0 = h'(c_1) = h'(c_2)$ for all $c_1 < x < c_2$, $h'(x) > 0$ for all $x > c_2$ or $x < c_1$, $h''(x) < 0 = h''(r)$ for all $x < r$, $h''(x) > 0$ for all $x > r$, $h(x)$ approaches to negative infinity at an exponential rate as $x \longrightarrow -\infty$, and $h(x)$ approaches to infinity at an exponential rate as $x \longrightarrow \infty$.



□

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Data availability

All data needed are contained in the manuscript.

References

[1] S.S. Almuthaybiri, C.C. Tisdell, Laminar flow in channels with porous walls: advancing the existence, uniqueness and approximation of solutions via fixed point approaches, J. Fixed Point Theory Appl. 24 (3) (2022) 55.
[2] A.S. Berman, Laminar flow in channels with porous walls, J. Appl. Phys. 24 (1953) 1232–1235.
[3] S.M. Cox, J.R. King, Large-Reynolds-number asymptotics of the Berman problem, Stud. Appl. Math. 113 (2004) 217–243.
[4] S.M. Cox, A.C. King, On the asymptotic solution of a high-order nonlinear ordinary differential equation, Proc. R. Soc. Lond. A Math. Phys. 453 (1997) 711–728.
[5] H. Guo, C. Gui, P. Lin, M. Zhao, Multiple solutions and their asymptotics for laminar flows through a porous channel with different permeabilities, IMA J. Appl. Math. 85 (2) (2020) 280–308.
[6] M.L. Hao, C.C. Tisdell, When is the porous, laminar flow problem with slip condition well posed? And where does the solution Lie?, Transp. Porous Media 147 (2) (2023) 281–303.
[7] C. Lu, On the asymptotic solution of laminar channel flow with large suction, SIAM J. Math. Anal. 28 (1997) 1113–1134.
[8] C. Lu, On matched asymptotic analysis for laminar channel flow with a turning point, Electron. J. Differ. Equ. 3 (1999) 109–118.
[9] C. Lu, A.D. MacGillivray, S.P. Hastings, Asymptotic behaviour of solutions of a similarity equation for laminar flows in channels with porous walls, IMA J. Appl. Math. 49 (1992) 139–162.
[10] A.D. MacGillivray, C. Lu, Asymptotic solution of a laminar flow in a porous channel with large suction: a nonlinear turning point problem, Methods Appl. Anal. 1 (1994) 229–248.
[11] J. Majdalani, D.S. Little, T.A. Jankowski, Vorticoacoustic waves in axisymmetric porous tubes with arbitrary suction, Phys. Fluids 37 (3) (2025).
[12] G. Raithby, Laminar heat transfer in the thermal entrance region of circular tubes and two-dimensional rectangular ducts with wall suction and injection, Int. J. Heat Mass Transf. 14 (1971) 223–243.
[13] W. Robinson, The existence of multiple solutions for the laminar flow in a uniformly porous channel with suction at both walls, J. Eng. Math. 10 (1976) 23–40.
[14] R. Terrill, Laminar flow in a uniformly porous channel with large injection, Aeronaut. Q. 16 (1965) 323–332.
[15] R. Terrill, G. Shrestha, Laminar flow through channels with porous walls and with an applied transverse magnetic field, Appl. Sci. Res. Sect. B 11 (1964) 134–144.
[16] M. Zaturska, P. Drazin, W. Banks, On the flow of a viscous fluid driven along a channel by suction at porous walls, Fluid Dyn. Res. 4 (1988) 151–178.